



E@B x Odys Final Presentation



Overview



● Methodology

Assignment Catchment Airports
Filtration of P2P and CE flyers

● Revenue Analysis

Pacific Northwest

SEA-Tac for Delta

Midwest

DTW & MSP for Delta

ORD for American

Houston Triangle

DFW for American

IAH for United

● Visualizations

● Helpful Functions

Total Revenue

Number of Drivers

Drivers Organized by Income Groups

Most Popular Tracts



Executive Summary

Objective: Determine revenue increase for airlines incorporating Odys eVTOLs by analyzing travel data and flight data to and from airports.

- The high number of catchment airports in the Midwest makes the ideal area to deploy Odys vehicles for both Delta and American Airlines.
- Delta could increase its annual revenue by 2.4% if Odys eVTOLs were integrated at MSP and DTW.
- American could increase its annual revenue by 1.6% if Odys eVTOLs were integrated at ORD.

Our Team



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Ethan Wang

Methodology



Catchment Airports by Hub



Primary Hub	Catchment Airports
SEA	BOI, GEG, PDX
ORD	CLE, CMH, CVG, DSM, DTW, GRR, IND, MCI, MDW, MSP, SDF, STL
DTW	BUF, CLE, CMH, CVG, GRR, MDW, ORD, IND, PIT, ROC, SDF
MSP	DSM, DTW, MCI, MDW, ORD
DFW	AMA, AUS, MAF, MEM, MSY, OKC, SAT, TUL
IAH	AMA, AUS, MAF, MEM, MSY, OKC, SAT, TUL

Passenger Selection Methodology



P2P

- Servicing drivers around the hub airport
- **Filter in: Drive to airport exceeds 40 mins (20 min drive to airport + 20 min wait for aircraft) + flight time**
 - To calculate flight time:
 $y = 1.8612E-07x^2 + 2.5885E-03x + 2.6245E-01$

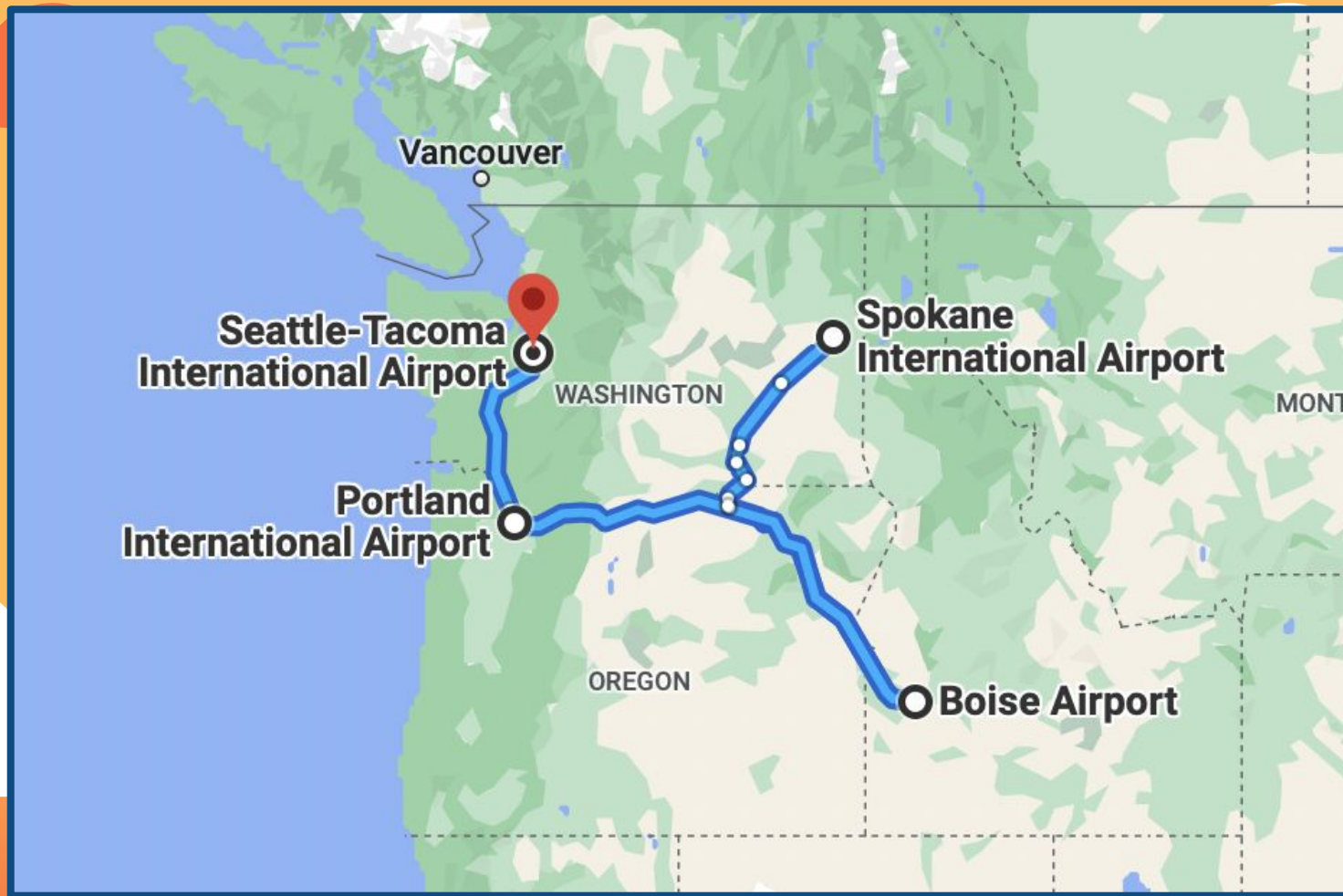
Catchment Expansion

- Servicing drivers to smaller, local airports and redirecting their traffic to hub airport
- **Filter in: Long haul flyers**
- **Filter in: Short haul flyers for whom it would be more convenient to fly to hub airport than drive to their local airport**
 - For both domestic and international flights:
 - Flights < 2000 miles $\Rightarrow$ short haul
 - Flights $\geq$ 2000 miles $\Rightarrow$ long haul



Delta Airlines

at SEA



SEA: Total Available Market



P2P: 60,902 drivers

from SEA area

CE: 48,679 drivers

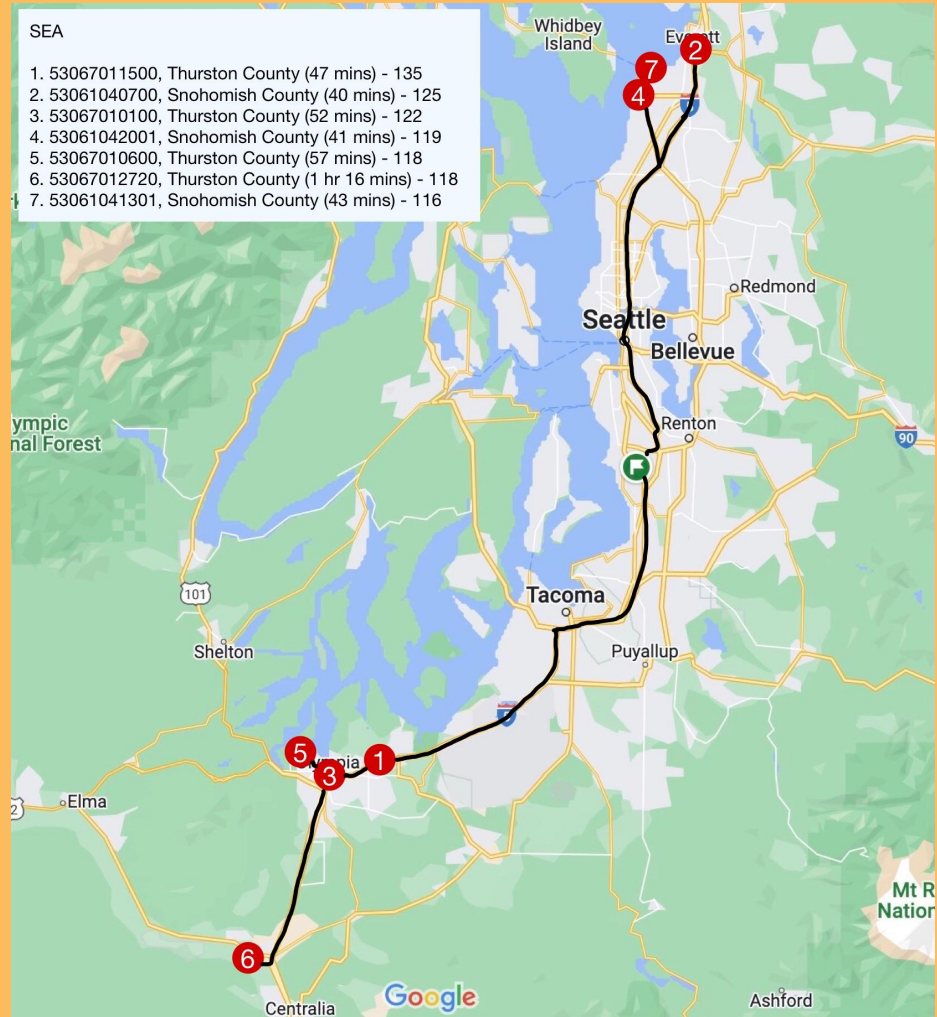
from BOI, GEG, PDX

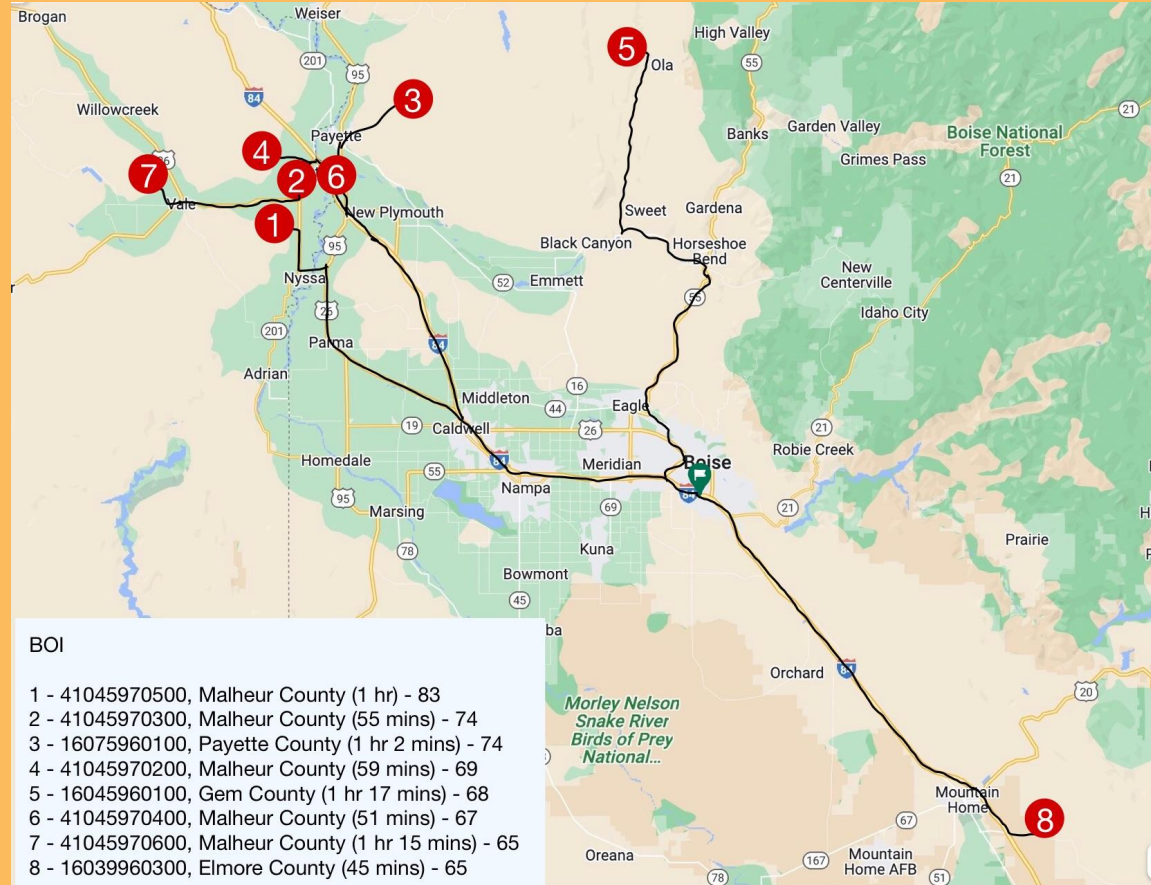




SEA

1. 53067011500, Thurston County (47 mins) - 135
2. 53061040700, Snohomish County (40 mins) - 125
3. 53067010100, Thurston County (52 mins) - 122
4. 53061042001, Snohomish County (41 mins) - 119
5. 53067010600, Thurston County (57 mins) - 118
6. 53067012720, Thurston County (1 hr 16 mins) - 118
7. 53061041301, Snohomish County (43 mins) - 116





SEA: Revenue from P2P and CE



P2P: \$7.13 M

60,902 drivers * \$117

CE: \$ 73.0 M

48,679 drivers * \$1500

└─ 88.7% short haul
11.3% long haul

Total: \$7.13 M + \$73.0 M = \$80.1 M



**Delta can increase annual
revenue by 0.2%
\$80.1 M / \$46.6 B[‡]
by integrating Odys
vehicles in SEA.**

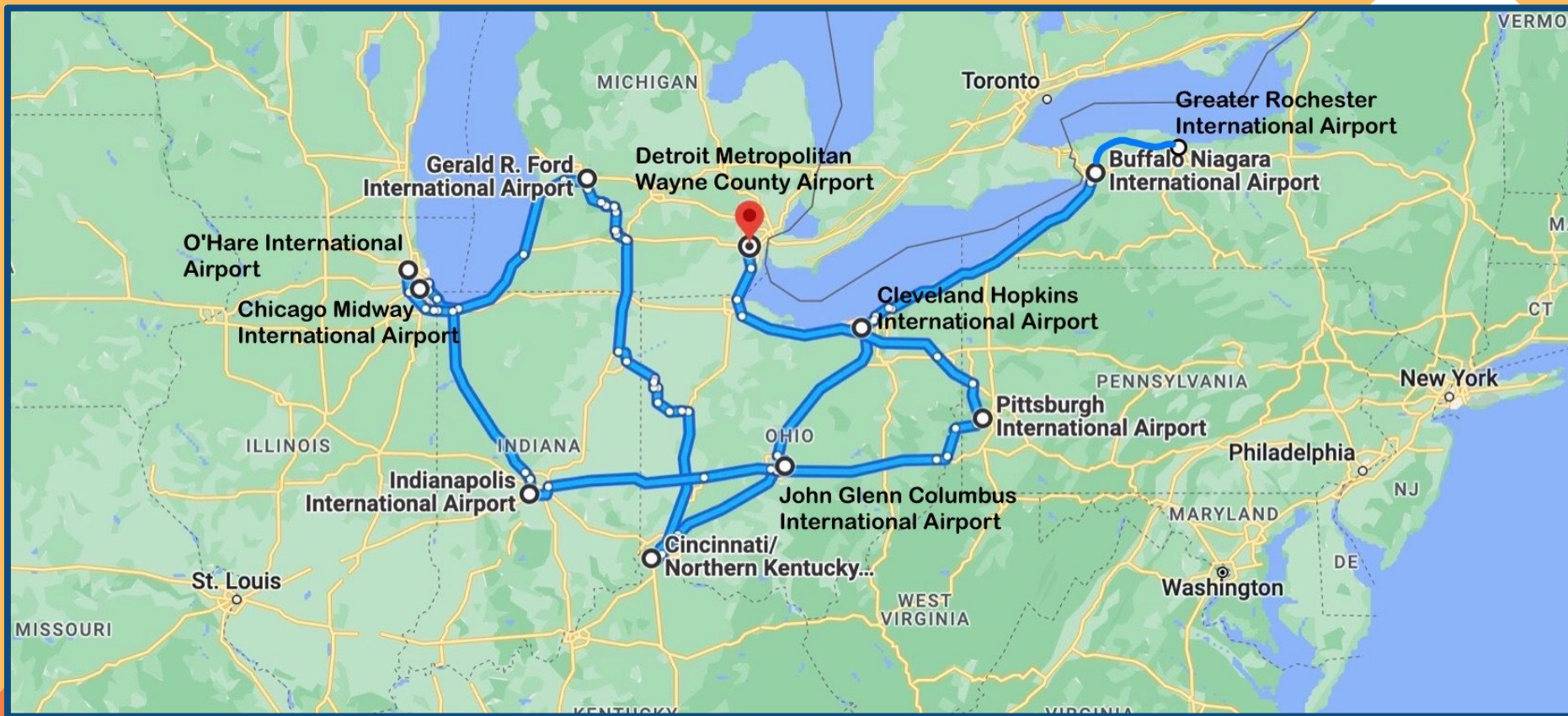
[‡]Last 12 months revenue, as of 9/30/2022





Delta Airlines

at DTW and MSP



DTW: Total Available Market



P2P: 96,576 drivers

from DTW area

CE: 506,519 drivers

from BUF, CLE, CVG, GRR,
IND, MDW, ORD, PIT, ROC,
SDF



DTW: Revenue from P2P and CE



P2P: \$11.3 M

96,576 drivers * \$117

CE: \$760 M

506,519 drivers * \$1500

└─ 94.8% short haul
5.2% long haul

Total: \$11.3 M + \$760 M = \$771 M





MSP: Total Available Market



P2P: 50,188 drivers

from MSP area

CE: 235,191 drivers

from DSM, DTW, MCI, MDW,
ORD



MSP: Revenue from P2P and CE



P2P: \$5.87 M

50,188 drivers * \$117

CE: \$353 M

235,191 drivers * \$1500

└─ 93.4% short haul
6.6% long haul

Total: \$5.87 M + \$353 M = \$359 M



**Delta can increase annual
revenue by 2.4%
(\$771 M + \$359 M) / \$46.6 B[‡]
by integrating Odys
vehicles in DTW & MSP.**

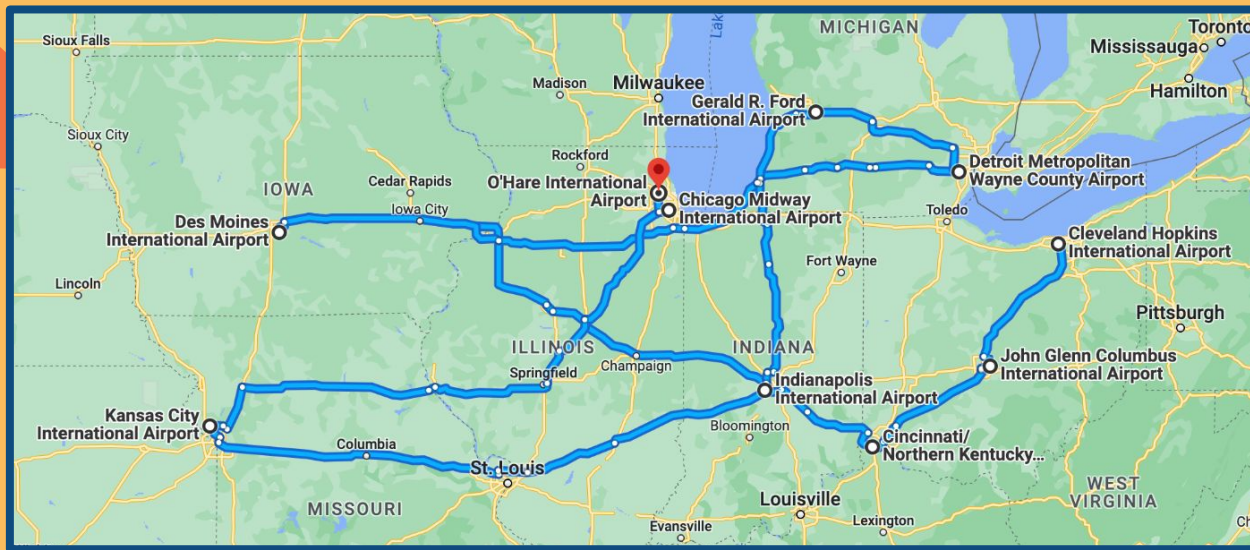
[‡]Last 12 months revenue, as of 9/30/2022





American Airlines

at ORD



ORD: Total Available Market



P2P: 163,508 drivers

from ORD area

CE: 473,555 drivers

from CLE, CMH, CVG, DSM,
DTW, GRR, IND, MCI, MDW,
MSP, SDF, STL



ORD: Revenue from P2P and CE



P2P: \$19.1 M

163,508 drivers * \$117

CE: \$710 M

473,555 drivers * \$1500

└─ 96.2% short haul
3.8% long haul

Total: \$19.1 M + \$710 M = \$729 M



**American Airlines can
increase annual revenue by
1.6%**

\$729 M / \$45.2 B[‡]

**by integrating Odys
vehicles in ORD.**

[‡]Last 12 months revenue, as of 9/30/2022





American Airlines

at DFW



DFW: Total Available Market



P2P: 56,298 drivers

from DFW area

CE: 80,522 drivers

from AMA, AUS, MAF, MEM,
MSY, OKC, SAT, TUL



DFW: Revenue from P2P and CE



P2P: \$6.59 M

56,298 drivers * \$117

CE: \$120 M

80,522 drivers * \$1500

└→ 98.6% short haul
1.4% long haul

Total: \$6.59 M + \$120 M = \$127 M



**American Airlines can
increase annual revenue by
.28%**

\$127 M / \$45.2 B[‡]

**by integrating Odys
vehicles in DFW.**

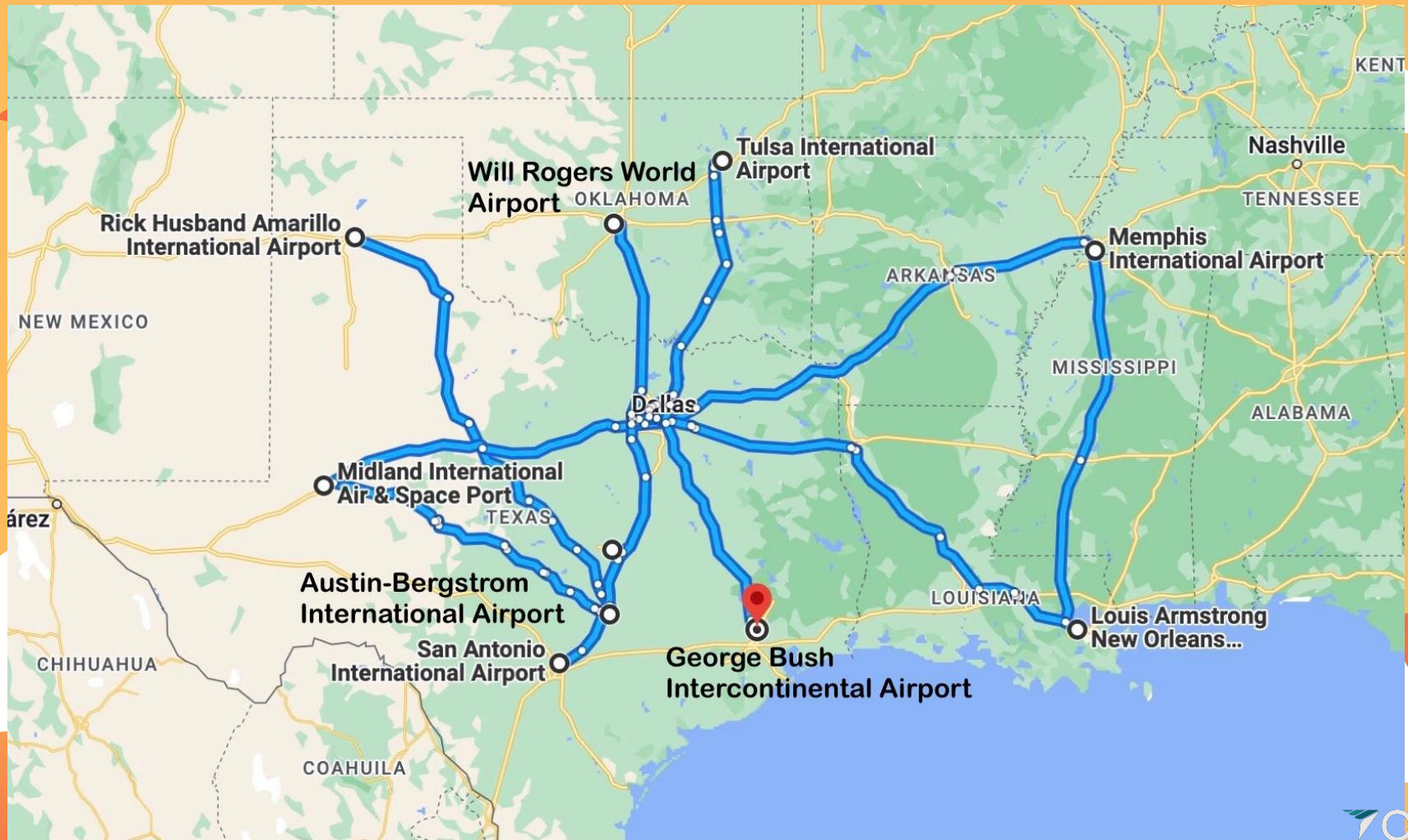
[‡]Last 12 months revenue, as of 9/30/2022





United Airlines

at IAH



IAH: Total Available Market



P2P: 45,086 drivers

from IAH area

CE: 75,863 drivers

from AMA, AUS, MAF, MEM,
MSY, OKC, SAT, TUL



IAH: Revenue from P2P and CE



P2P: \$5.3 M

45,086 drivers * \$117

CE: \$114 M

75,863 drivers * \$1500

└─ 98.6% short haul
1.4% long haul

Total: \$5.3M + \$114 M = \$119 M

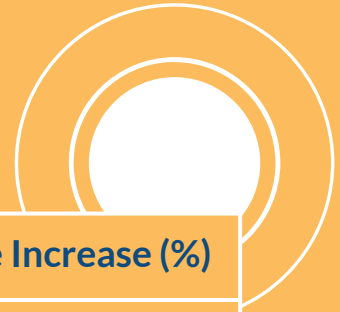


**United Airlines can increase
annual revenue by **.29%**
\$119 M / \$40.7 B[‡]
by integrating Odys
vehicles in IAH.**

[‡]Last 12 months revenue, as of 9/30/2022



Total Revenue Increases



		Number of Flyers	Total Revenue (\$)	Revenue Increase (%)
Delta	SEA	109,581	80.1 M	.20
Delta	MSP	285,379	359 M	2.4
Delta	DTW	603,095	771 M	
American	ORD	637,063	729 M	1.6
American	DFW	136,820	127 M	.28
United	IAH	120,949	119 M	.29



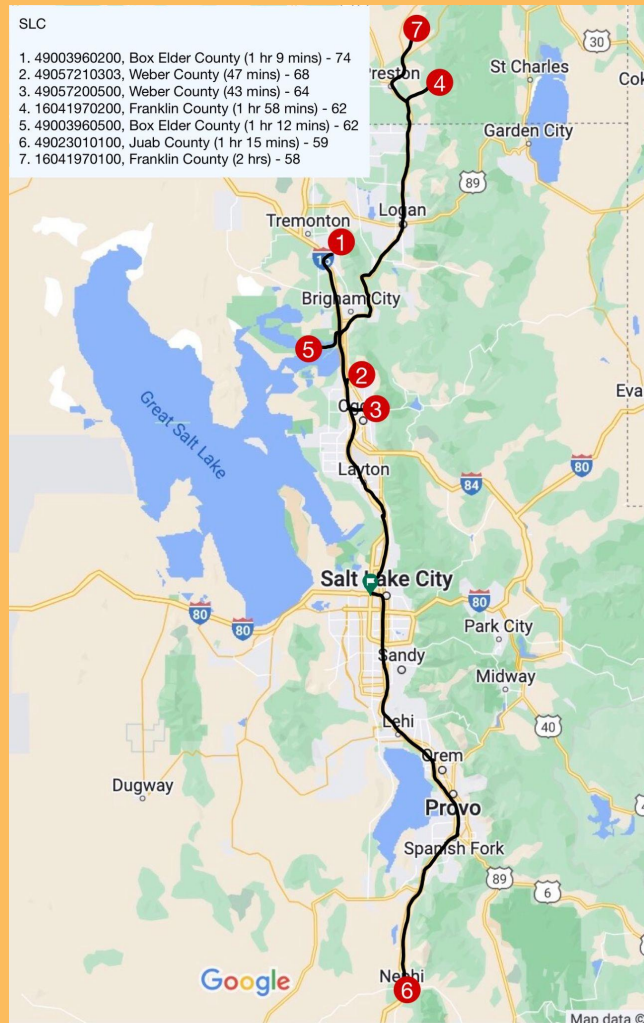
Appendix

West Coast



SLC

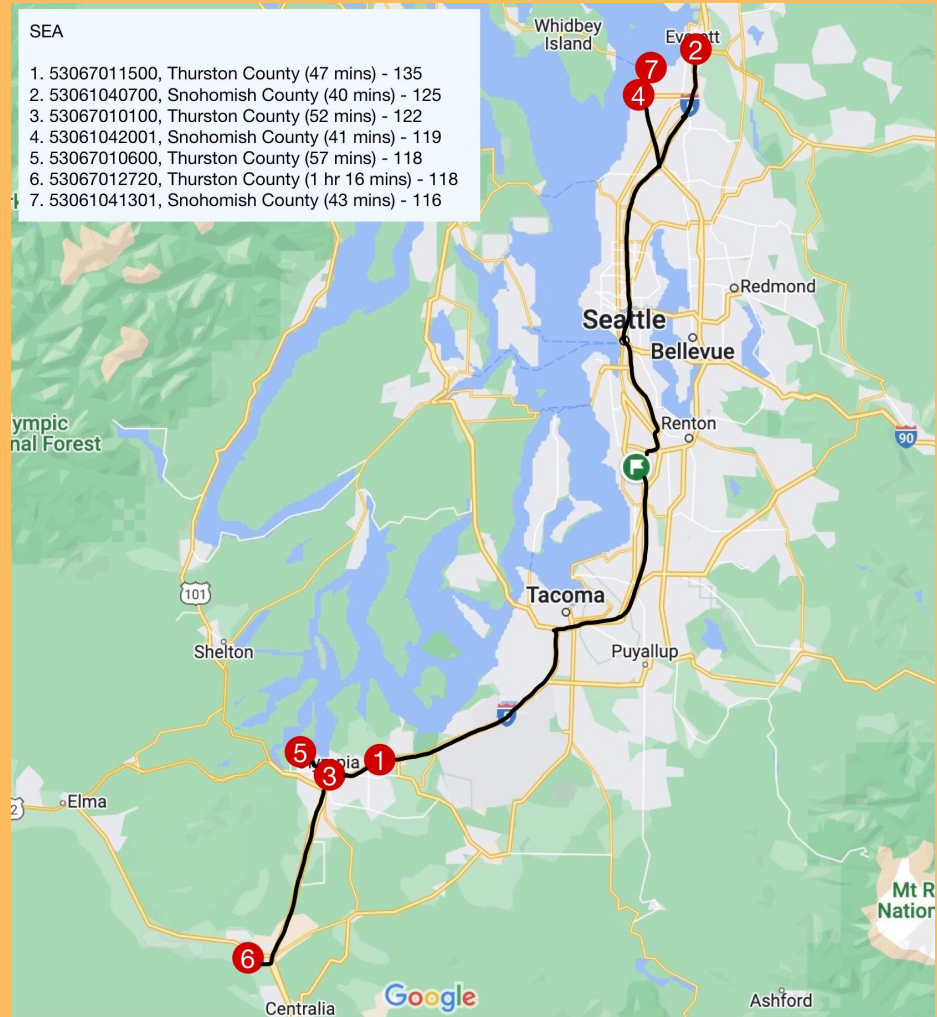
1. 49003960200, Box Elder County (1 hr 9 mins) - 74
2. 49057210303, Weber County (47 mins) - 68
3. 49057200500, Weber County (43 mins) - 64
4. 16041970200, Franklin County (1 hr 58 mins) - 62
5. 49003960500, Box Elder County (1 hr 12 mins) - 62
6. 49023010100, Juab County (1 hr 15 mins) - 59
7. 16041970100, Franklin County (2 hrs) - 58

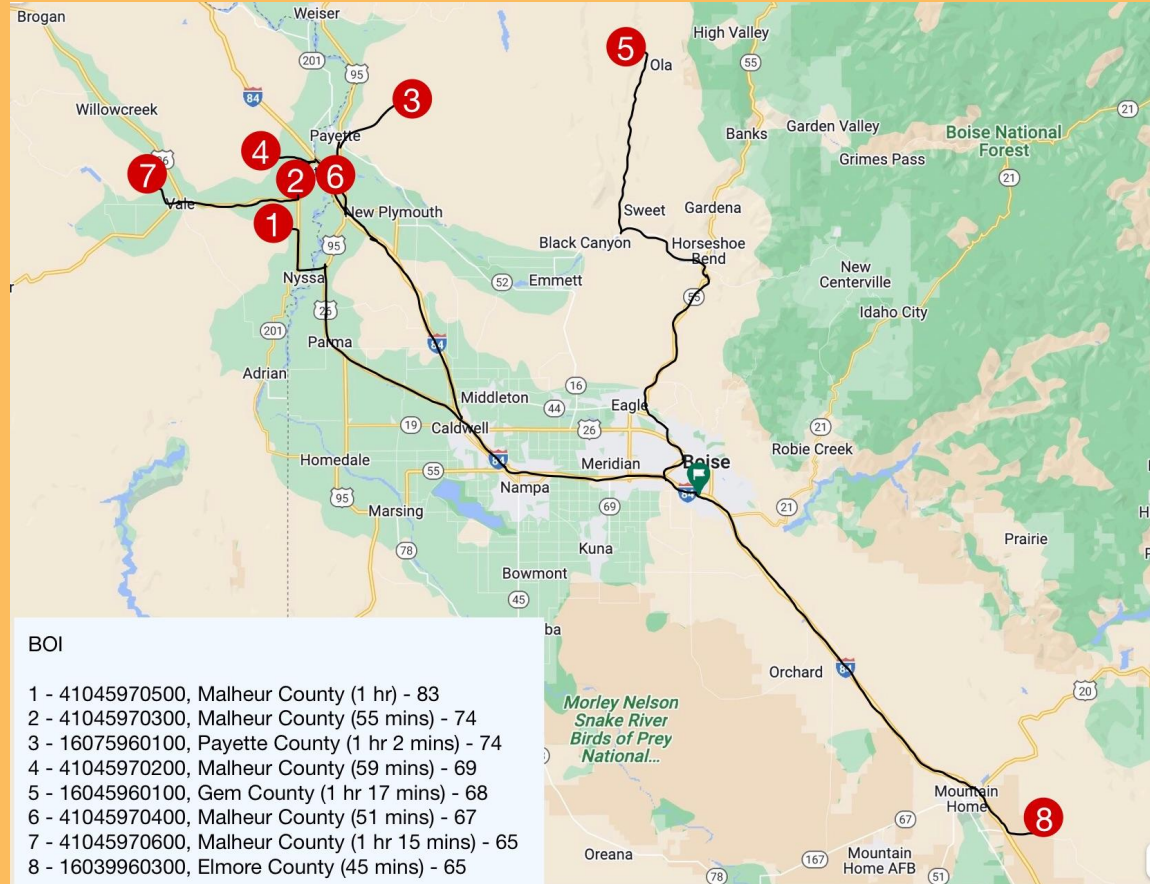


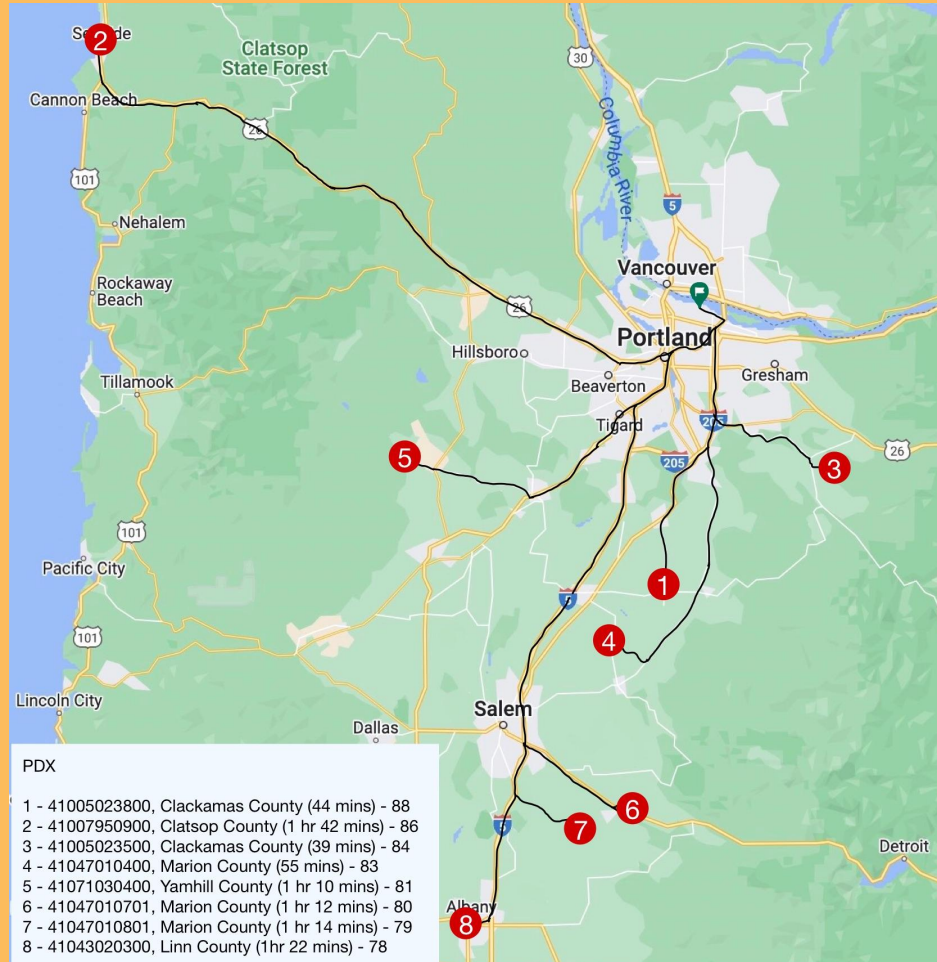


SEA

1. 53067011500, Thurston County (47 mins) - 135
2. 53061040700, Snohomish County (40 mins) - 125
3. 53067010100, Thurston County (52 mins) - 122
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6. 53067012720, Thurston County (1 hr 16 mins) - 118
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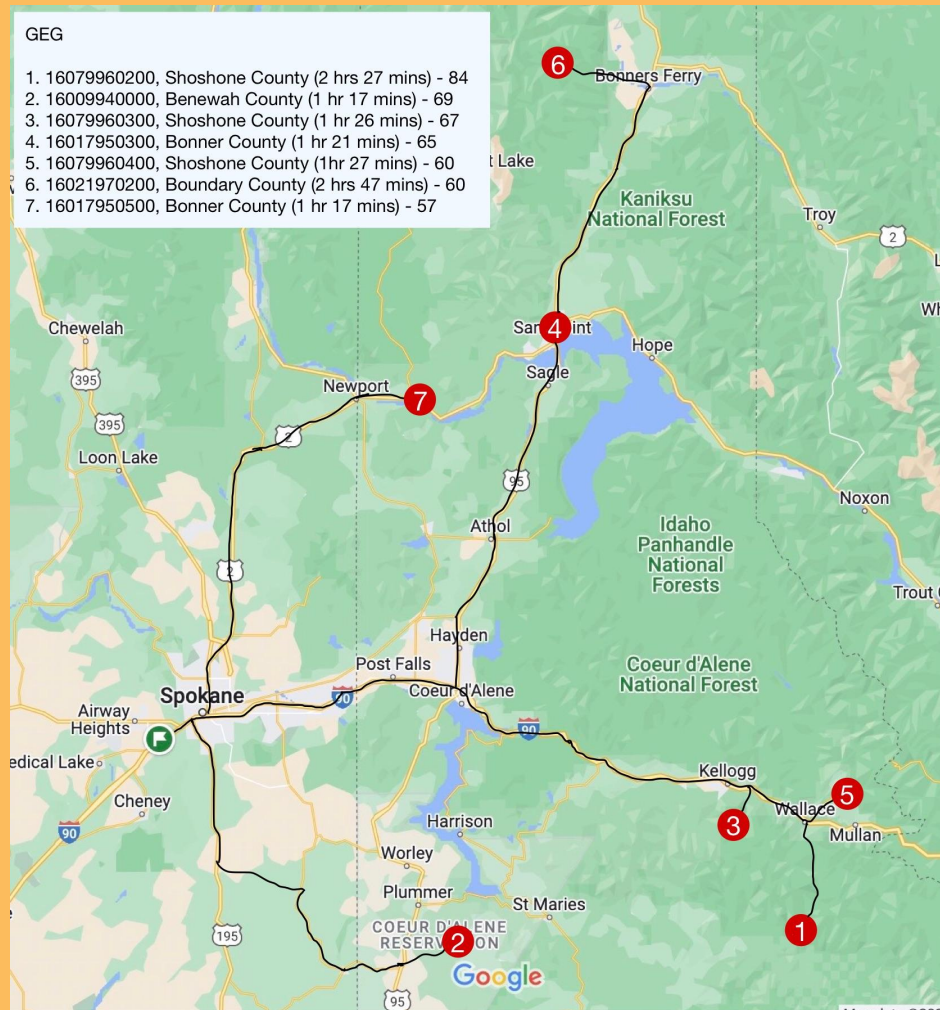




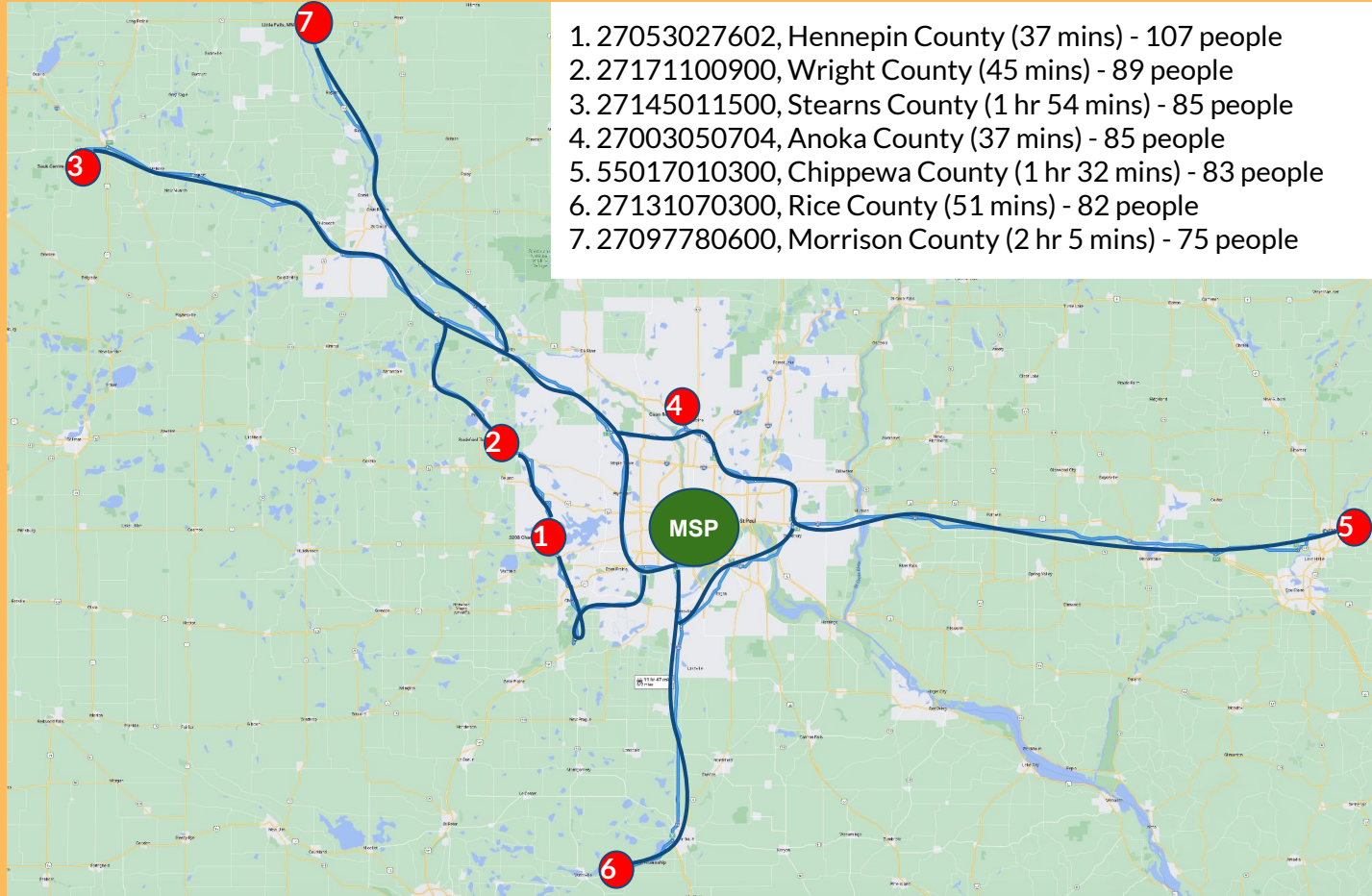


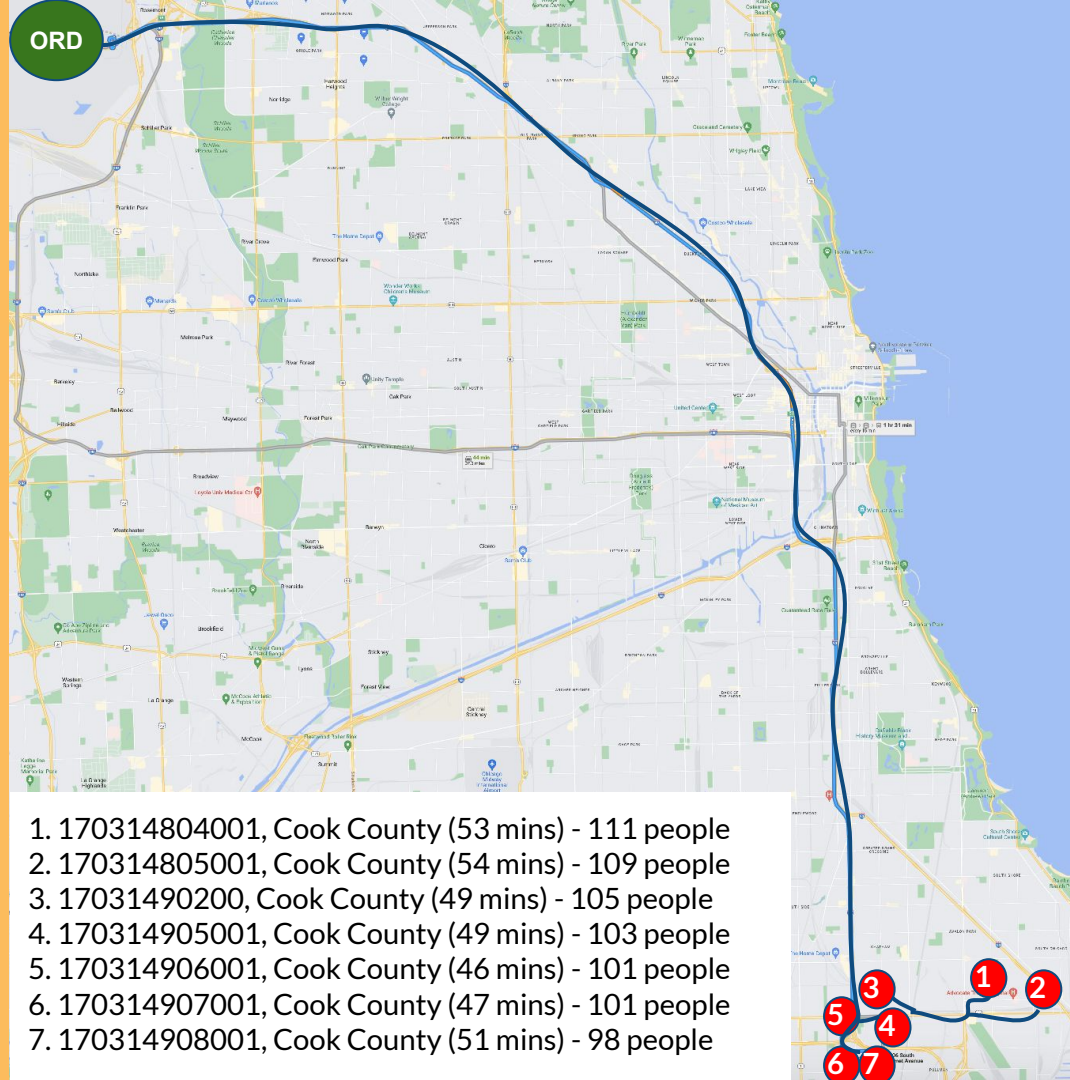
GEG

1. 16079960200, Shoshone County (2 hrs 27 mins) - 84
2. 16009940000, Benewah County (1 hr 17 mins) - 69
3. 16079960300, Shoshone County (1 hr 26 mins) - 67
4. 16017950300, Bonner County (1 hr 21 mins) - 65
5. 16079960400, Shoshone County (1hr 27 mins) - 60
6. 16021970200, Boundary County (2 hrs 47 mins) - 60
7. 16017950500, Bonner County (1 hr 17 mins) - 57



Midwest

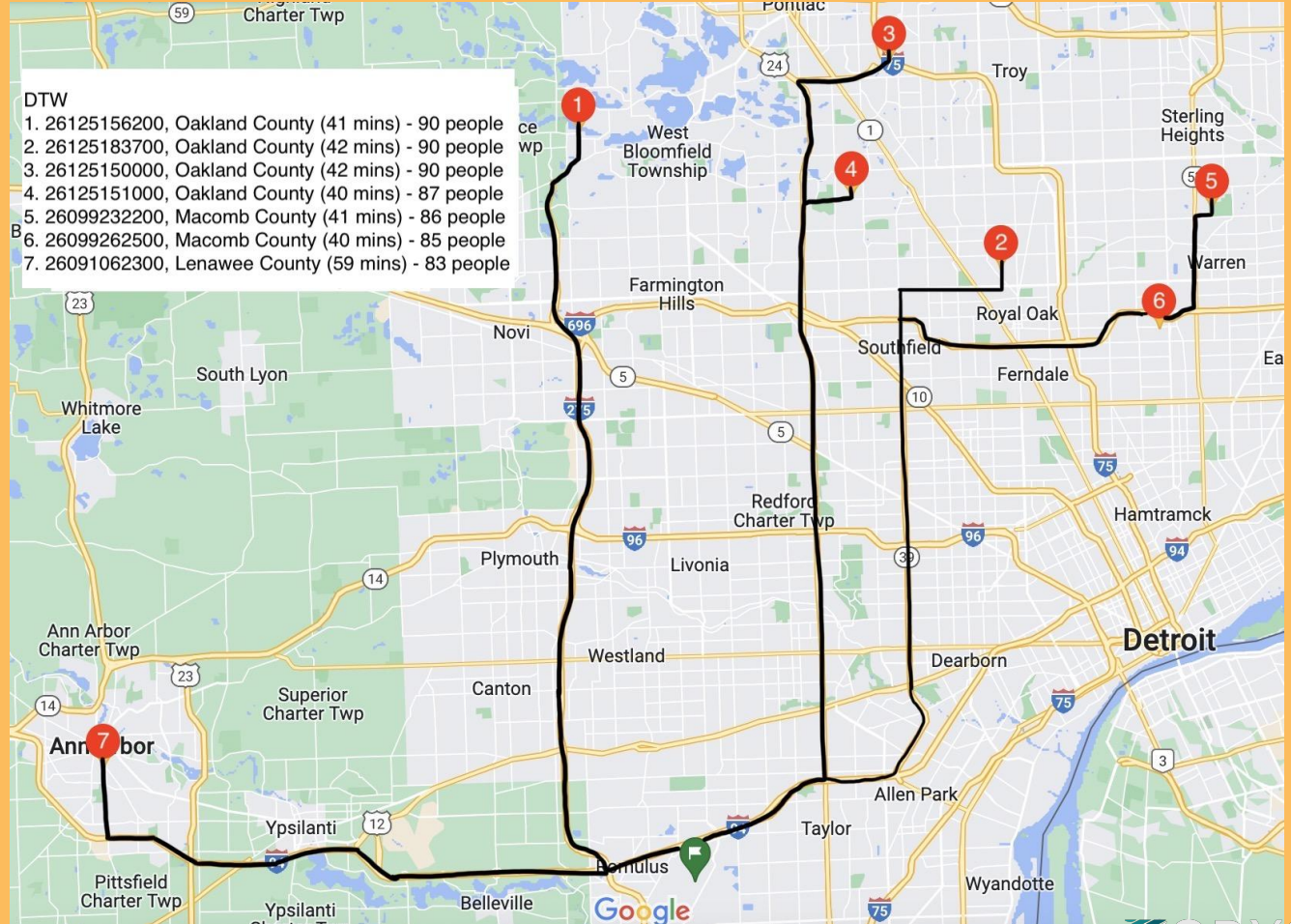






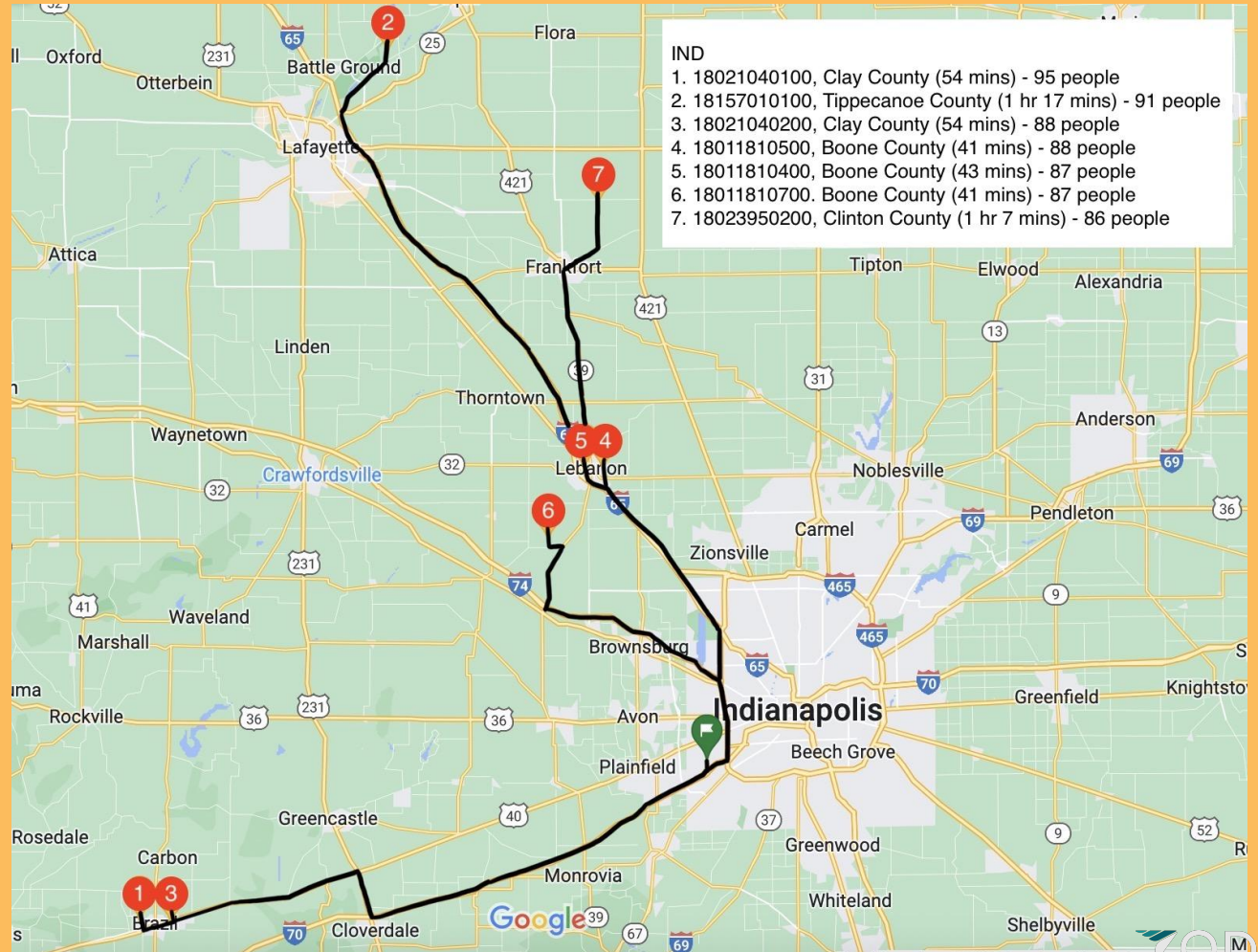
DTW

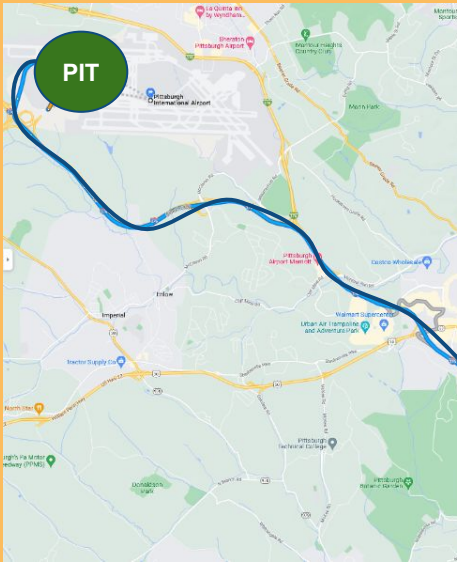
1. 26125156200, Oakland County (41 mins) - 90 people
2. 26125183700, Oakland County (42 mins) - 90 people
3. 26125150000, Oakland County (42 mins) - 90 people
4. 26125151000, Oakland County (40 mins) - 87 people
5. 26099232200, Macomb County (41 mins) - 86 people
6. 26099262500, Macomb County (40 mins) - 85 people
7. 26091062300, Lenawee County (59 mins) - 83 people



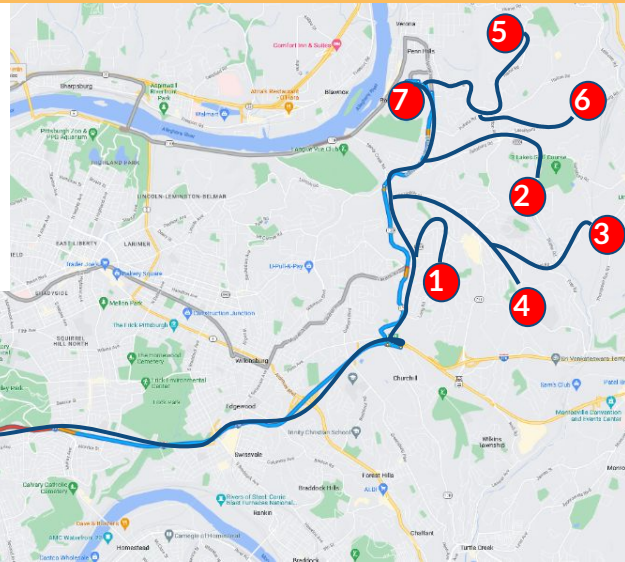
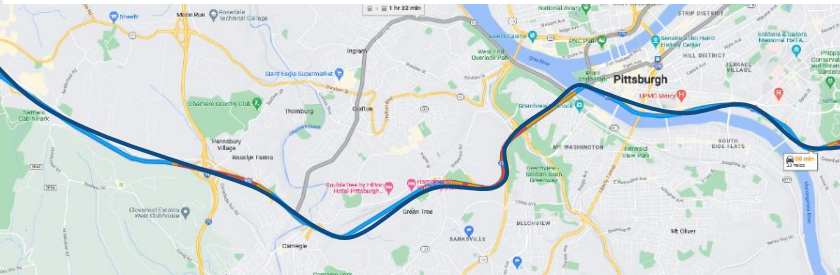


IND



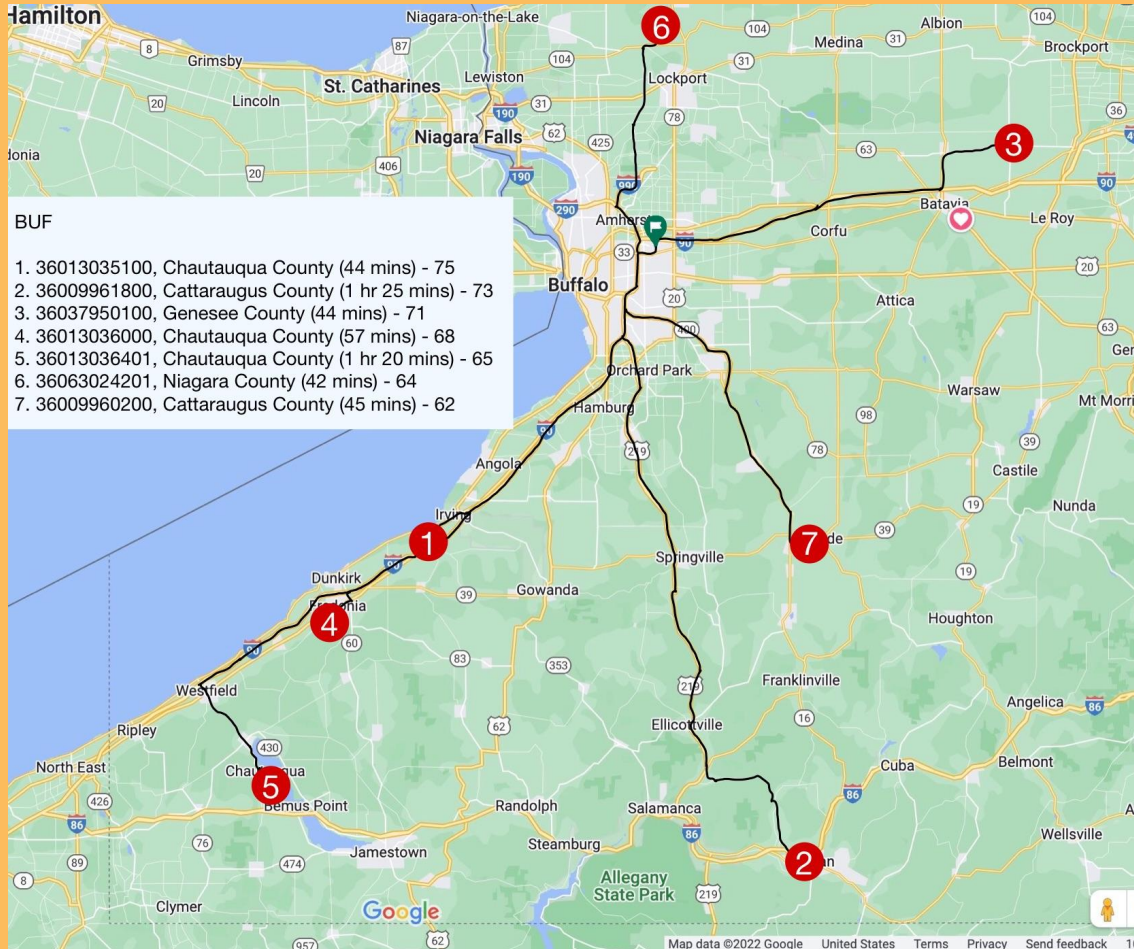


1. 420035234001, Allegheny County (35 mins) - 103 people
2. 420035235011, Allegheny County (39 mins) - 103 people
3. 420035235021, Allegheny County (38 mins) - 91 people
4. 420035236001, Allegheny County (37 mins) - 90 people
5. 420035237011, Allegheny County (44 mins) - 90 people
6. 420035237021, Allegheny County (43 mins) - 89 people
7. 420035238001, Allegheny County (42 mins) - 87 people



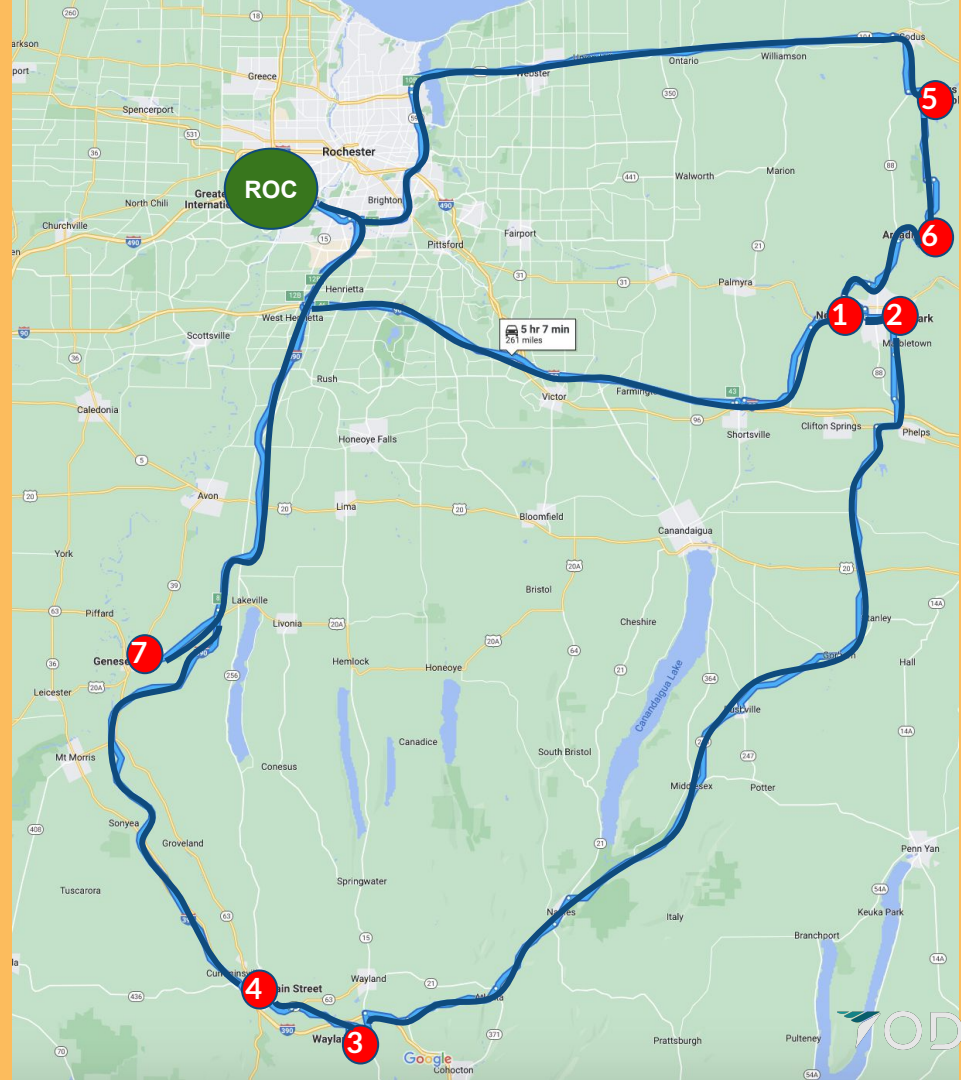


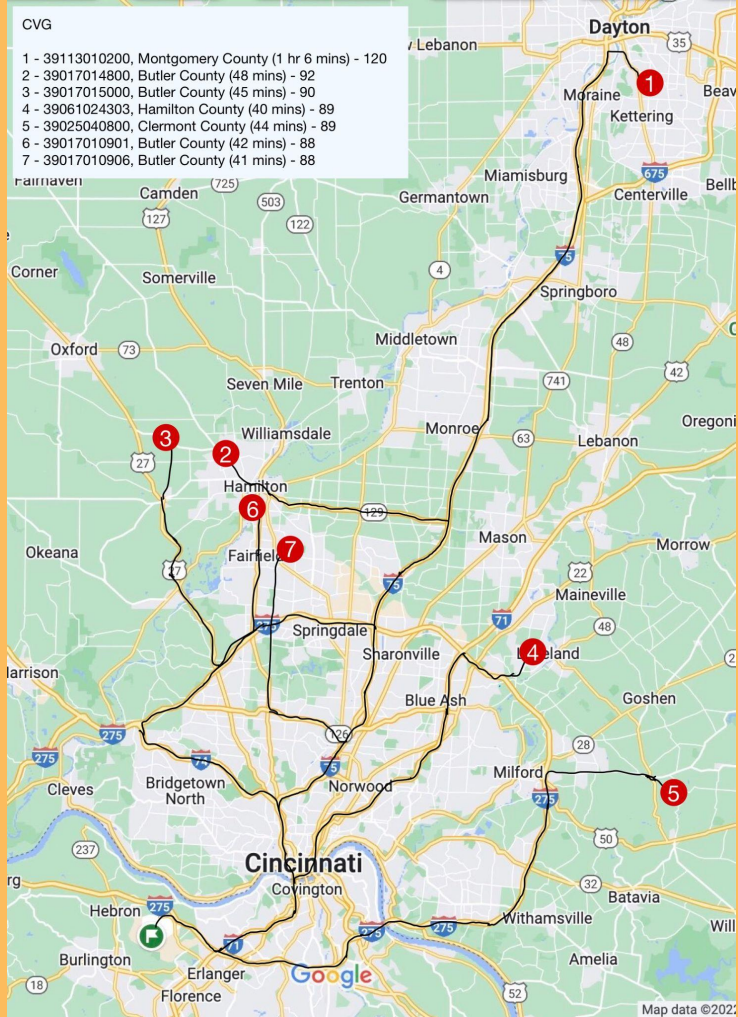
BUF





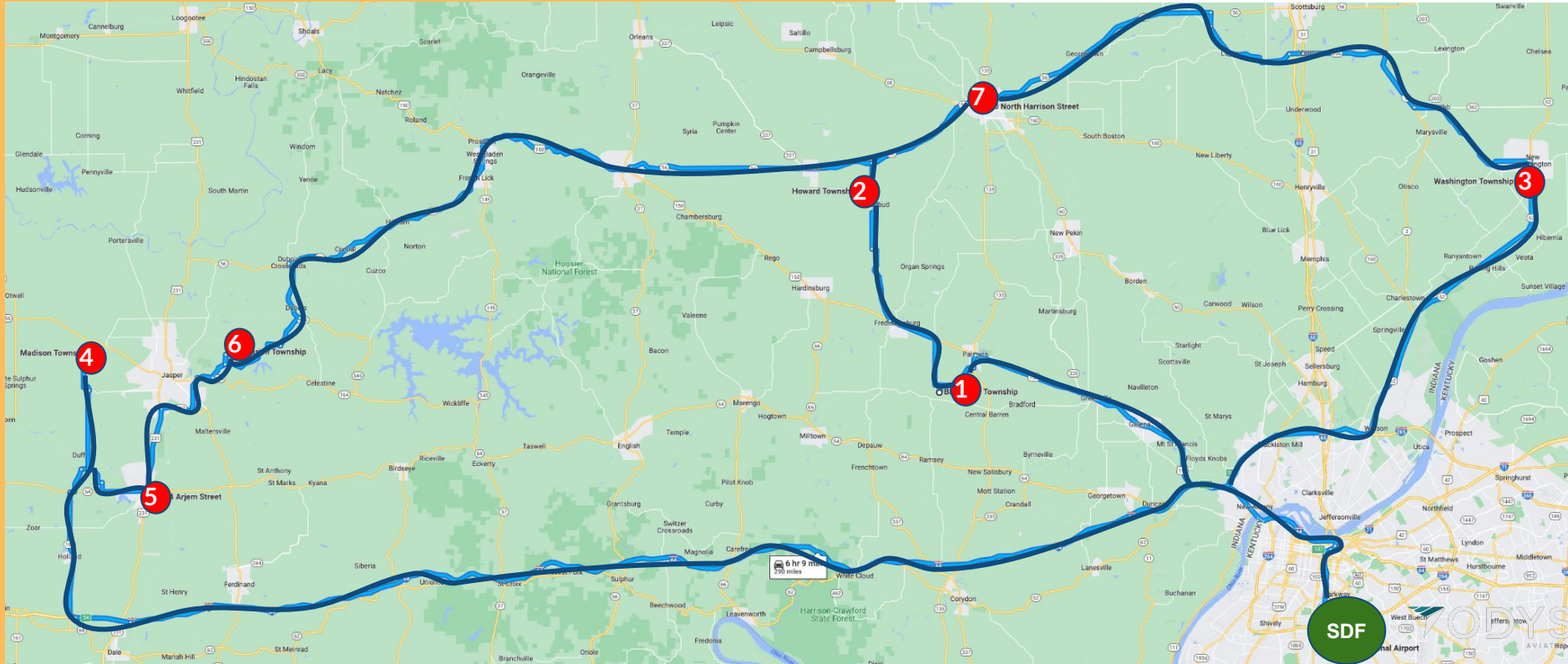
1. 36117021100, Wayne County (47 mins) - 65 people
2. 36117021200, Wayne County (50 mins) - 61 people
3. 36101960400, Steuben County (1 hr) - 58 people
4. 36051031400, Livingston County (53 mins) - 55 people
5. 36117020900, Wayne County (48 mins) - 54 people
6. 36117021000, Wayne County (53 mins) - 54 people
7. 36051030400, Livingston County (36 mins) - 53 people

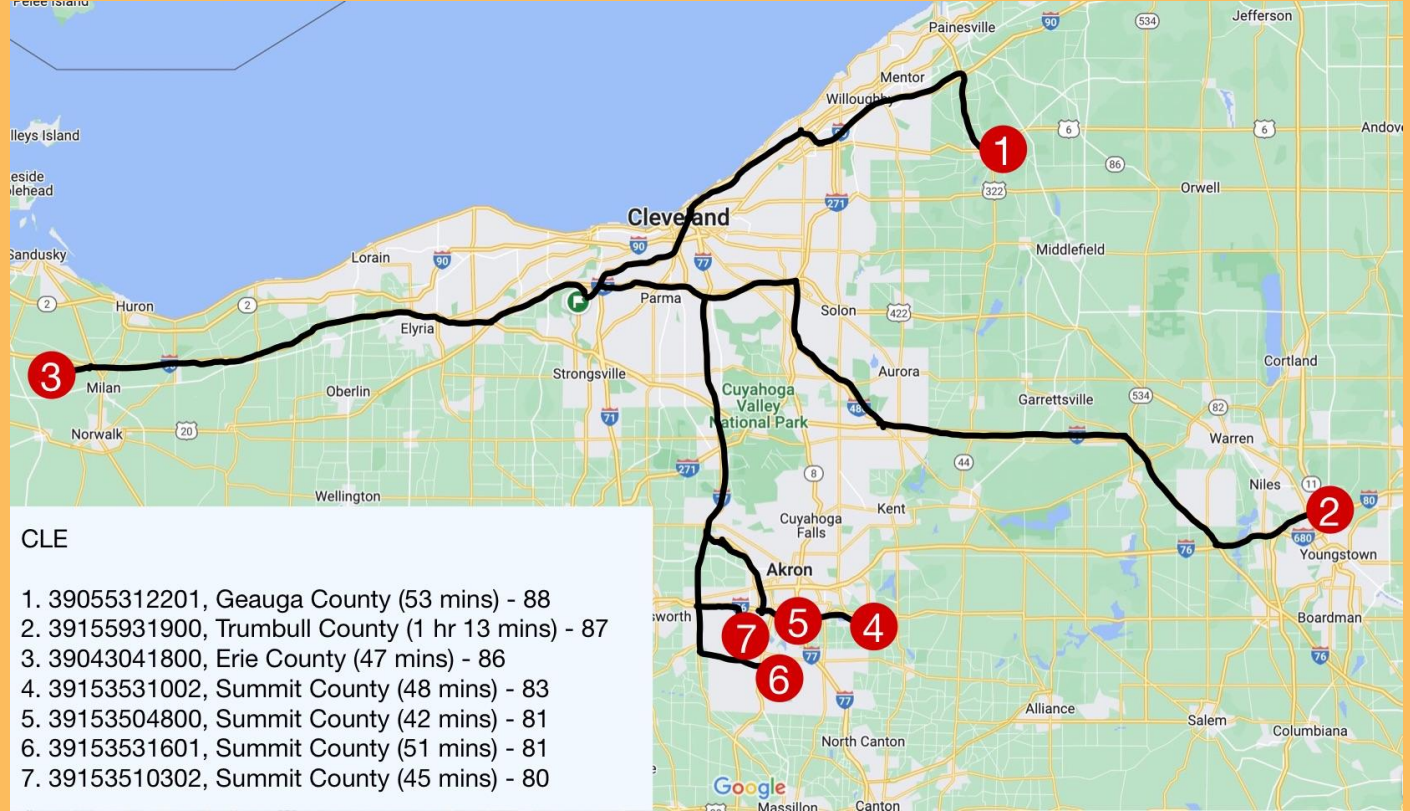






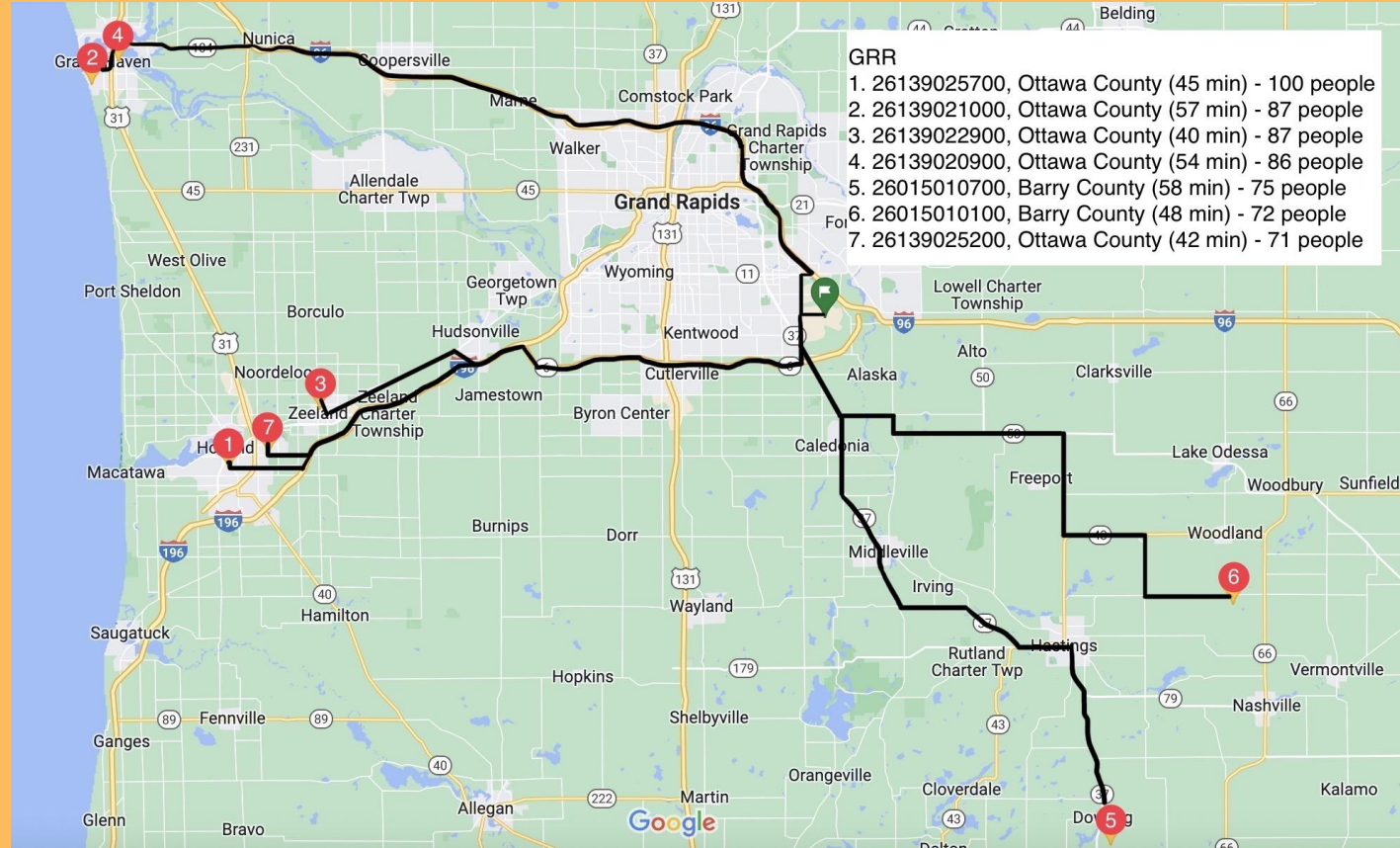
1. 18061060100, Harrison County (45 mins) - 68 people
2. 18175967400, Washington County (57 mins) - 65 people
3. 18019051000, Clark County (42 mins)- 64 people
4. 18037953200, Dubois County (1 hr 33 mins) - 64 people
5. 18037953800, Dubois County (1 hr 20 mins) - 62 people
6. 18037953500, Dubois County (1 hr 28 mins) - 62 people
7. 18175967500, Washington County (53 mins) - 60 people

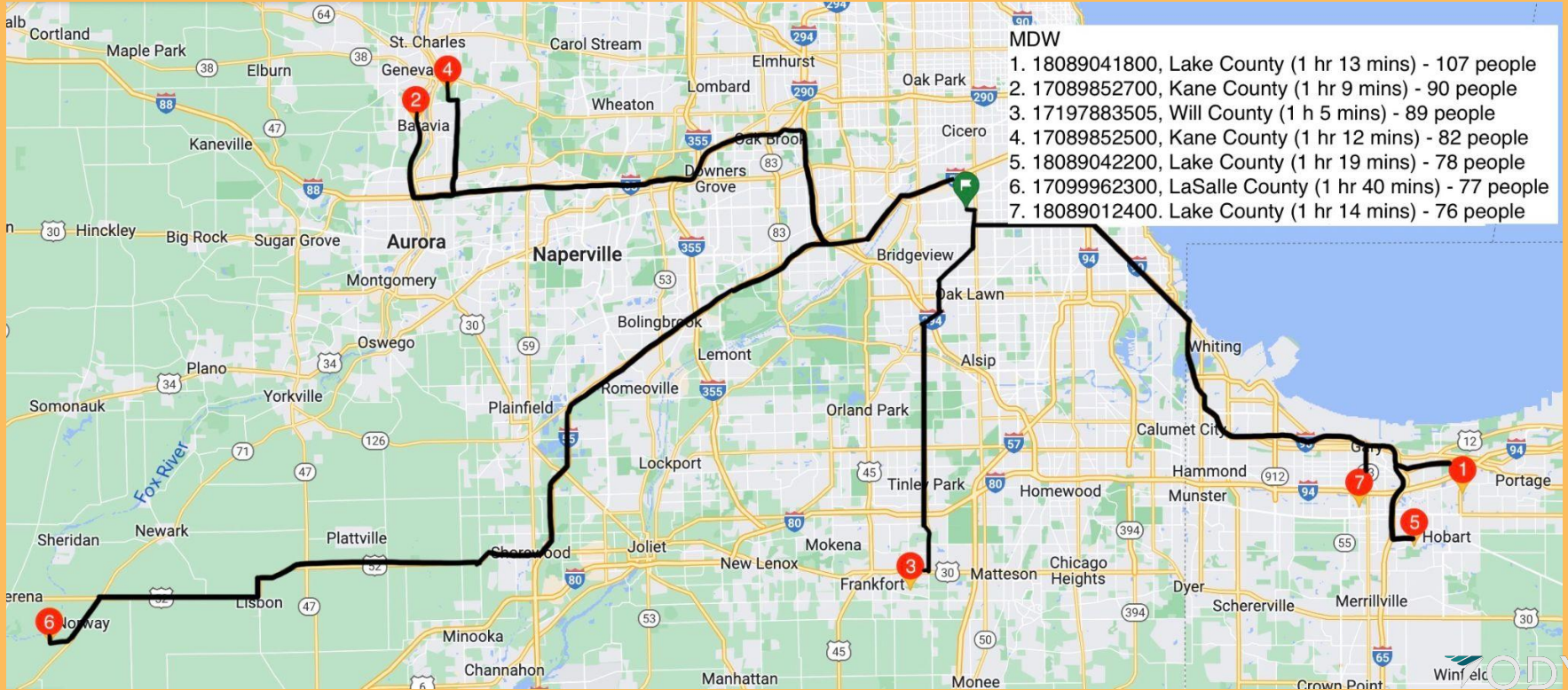






GRR

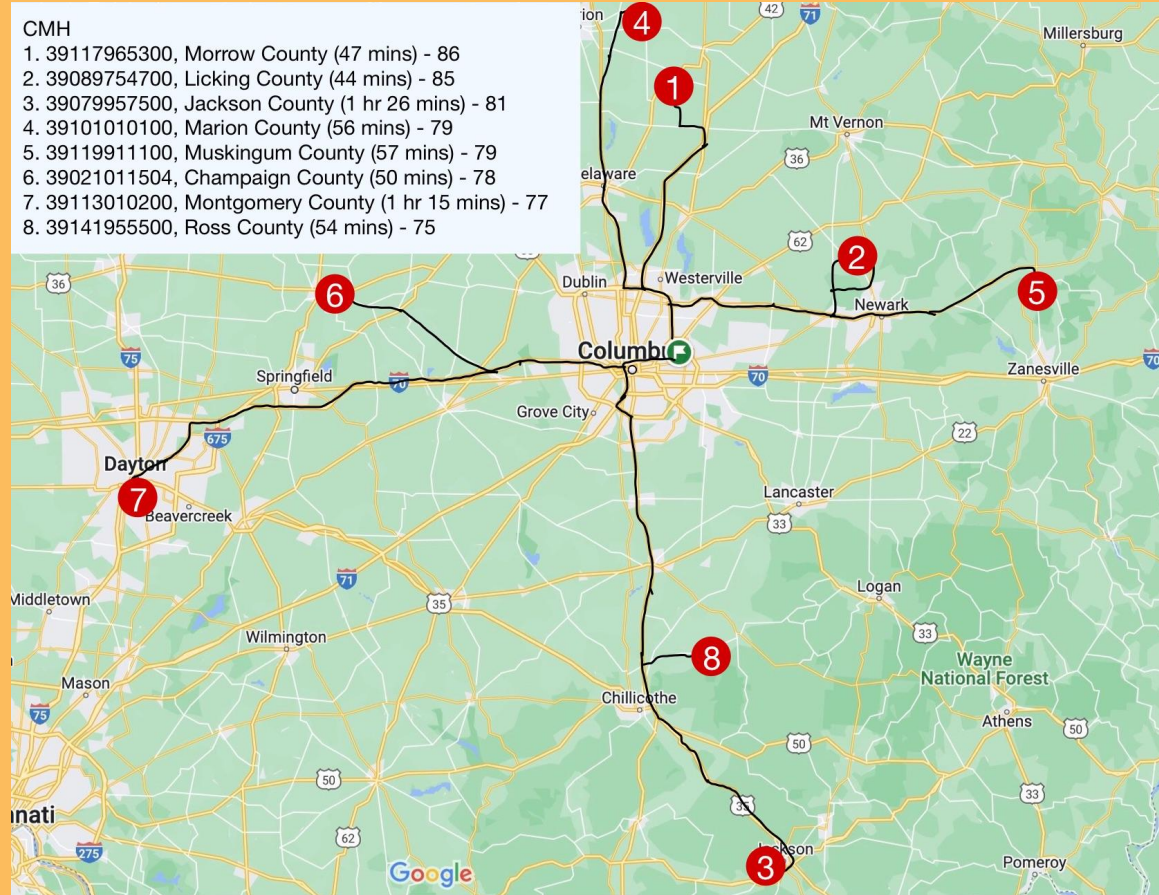






CMH

1. 39117965300, Morrow County (47 mins) - 86
2. 39089754700, Licking County (44 mins) - 85
3. 39079957500, Jackson County (1 hr 26 mins) - 81
4. 39101010100, Marion County (56 mins) - 79
5. 39119911100, Muskingum County (57 mins) - 79
6. 39021011504, Champaign County (50 mins) - 78
7. 39113010200, Montgomery County (1 hr 15 mins) - 77
8. 39141955500, Ross County (54 mins) - 75



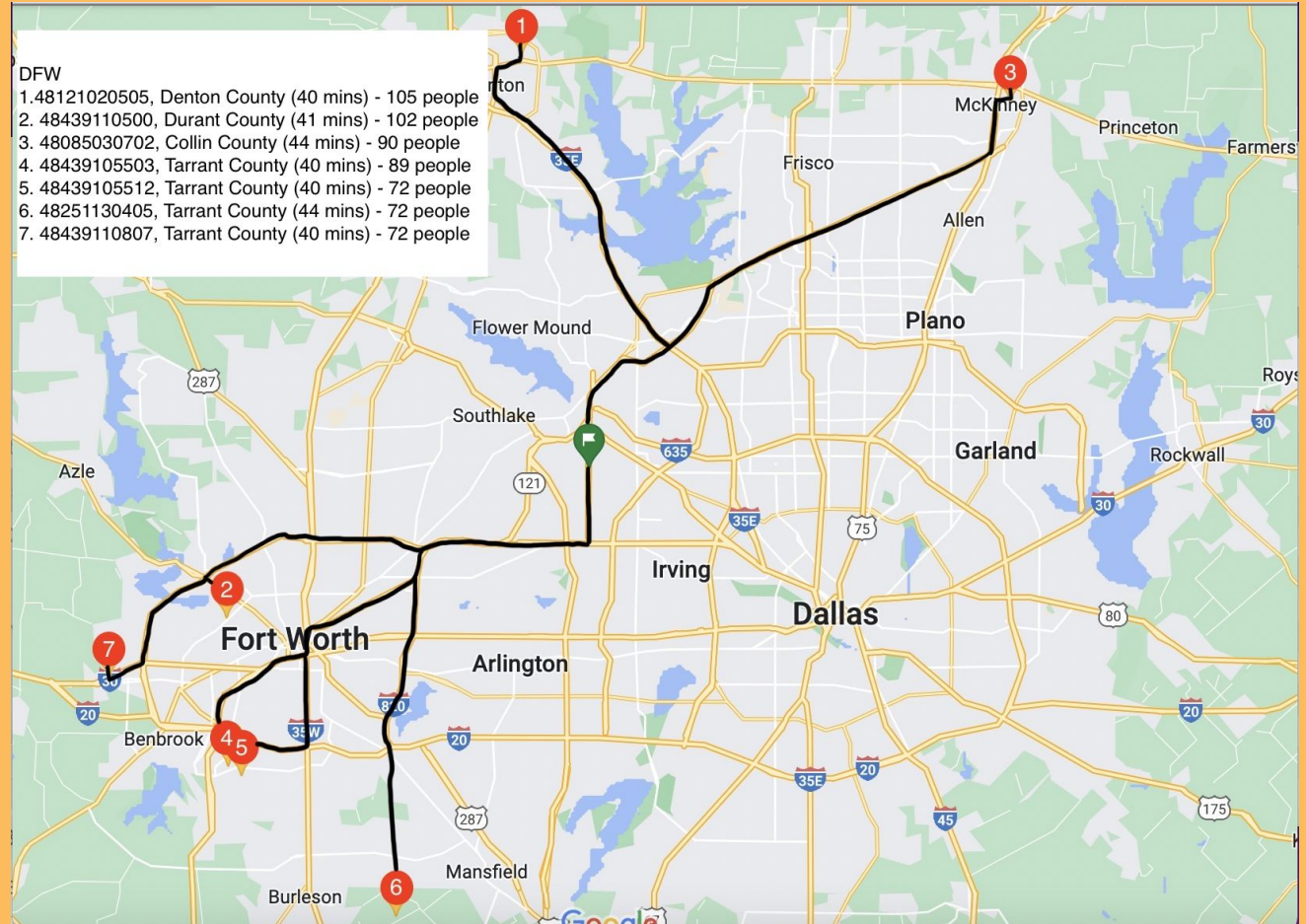
Texas Triangle



DFW

DFW

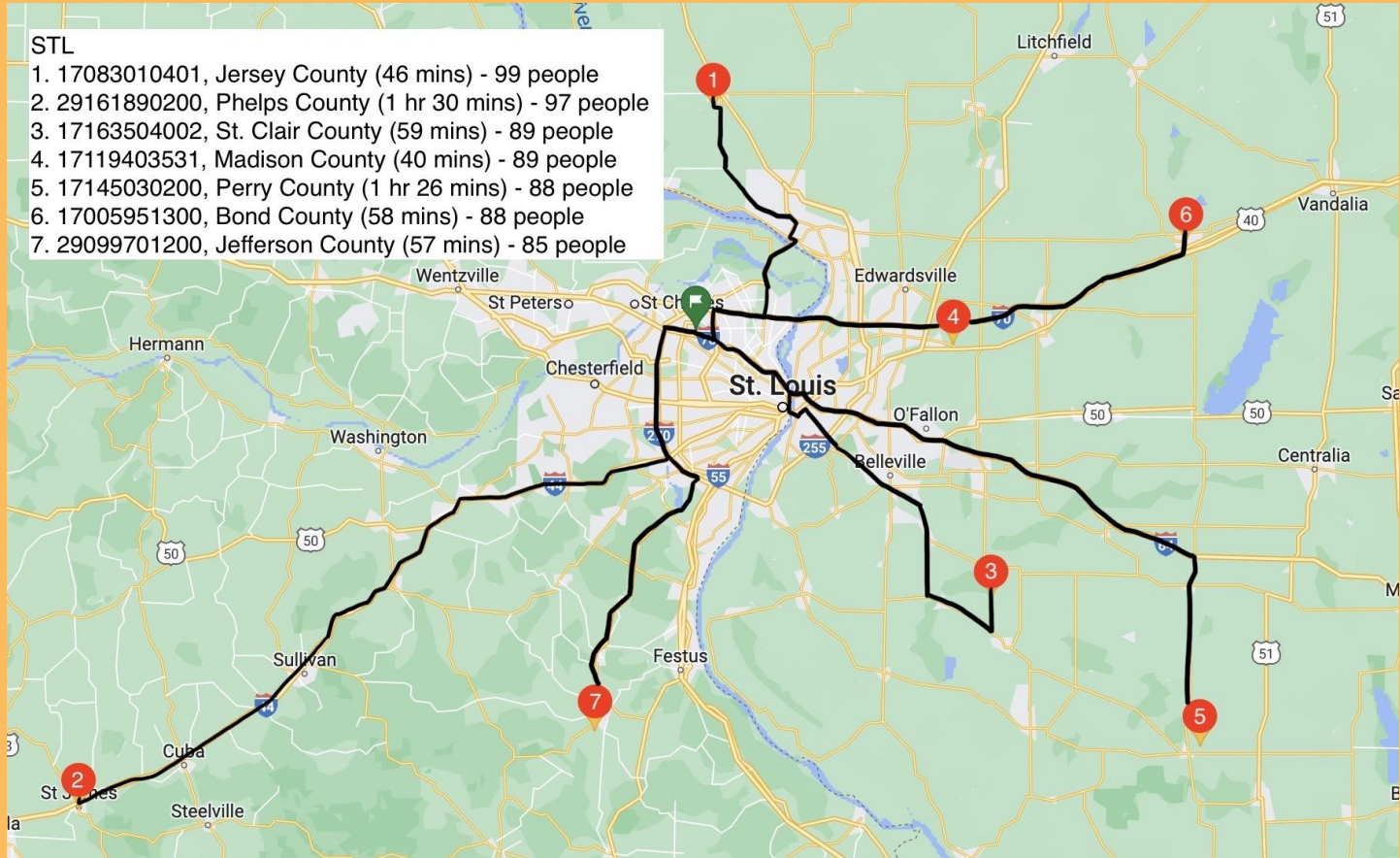
1. 48121020505, Denton County (40 mins) - 105 people
2. 48439110500, Durant County (41 mins) - 102 people
3. 48085030702, Collin County (44 mins) - 90 people
4. 48439105503, Tarrant County (40 mins) - 89 people
5. 48439105512, Tarrant County (40 mins) - 72 people
6. 48251130405, Tarrant County (44 mins) - 72 people
7. 48439110807, Tarrant County (40 mins) - 72 people

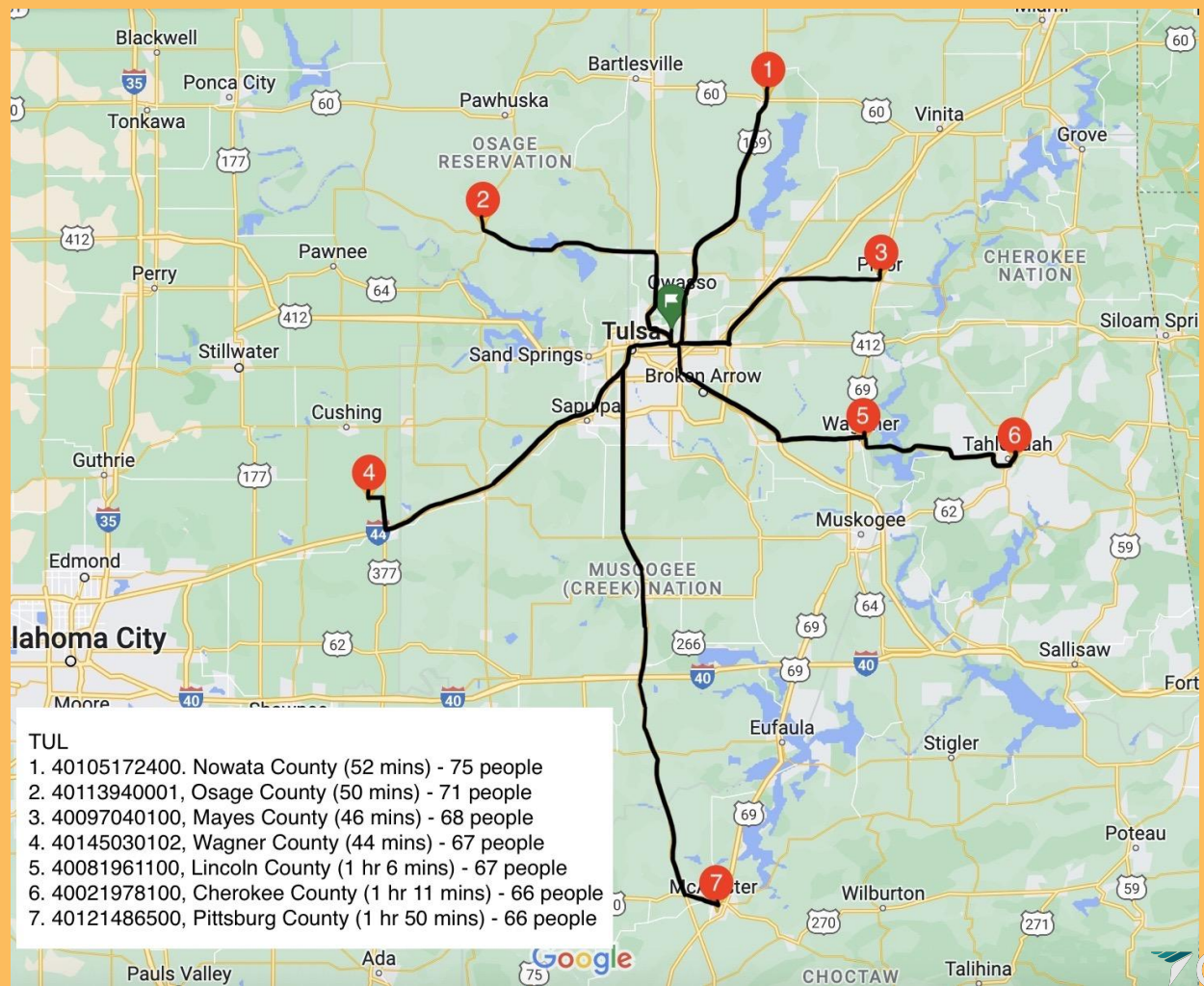




STL

1. 17083010401, Jersey County (46 mins) - 99 people
2. 29161890200, Phelps County (1 hr 30 mins) - 97 people
3. 17163504002, St. Clair County (59 mins) - 89 people
4. 17119403531, Madison County (40 mins) - 89 people
5. 17145030200, Perry County (1 hr 26 mins) - 88 people
6. 17005951300, Bond County (58 mins) - 88 people
7. 29099701200, Jefferson County (57 mins) - 85 people



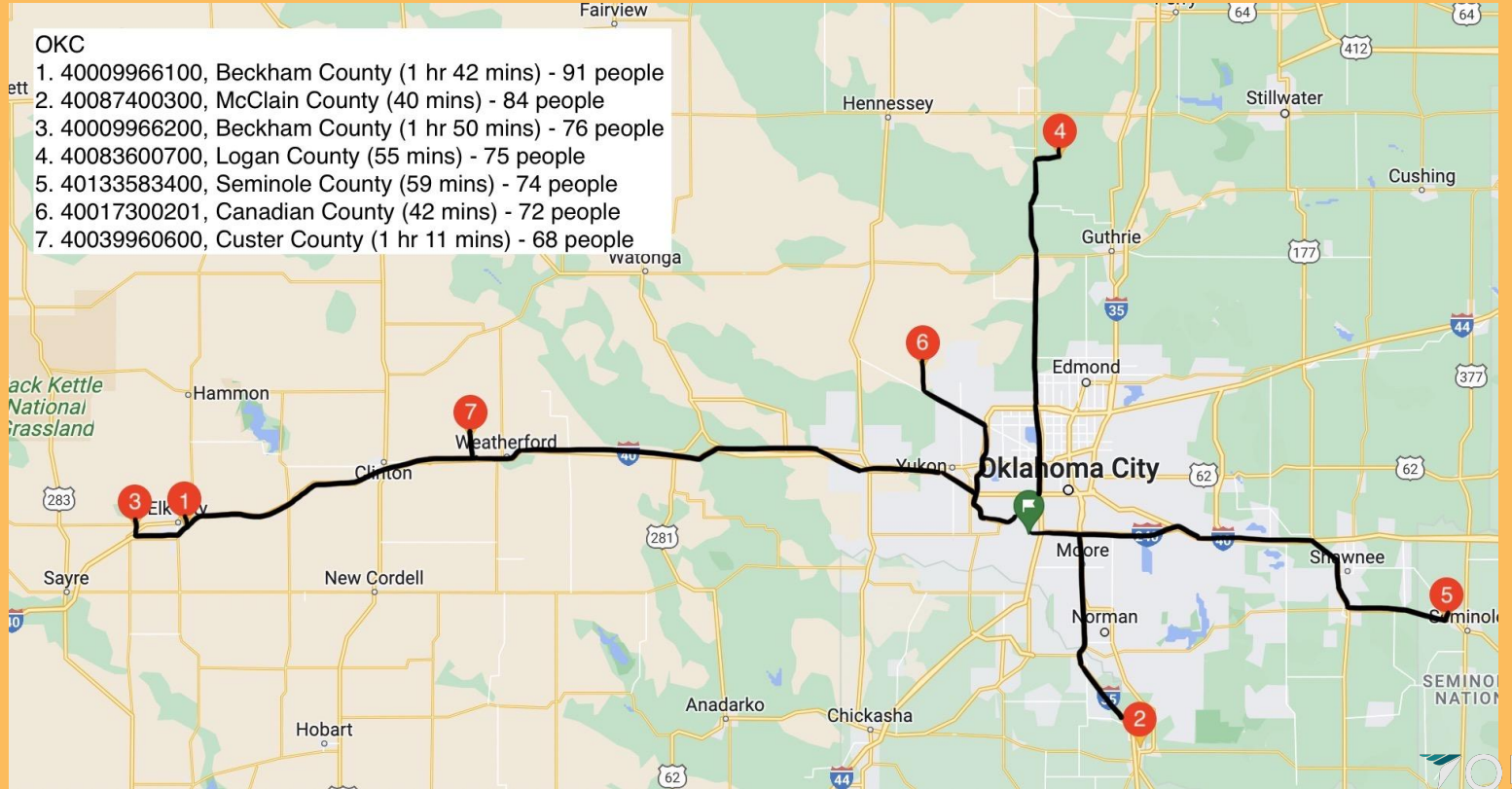


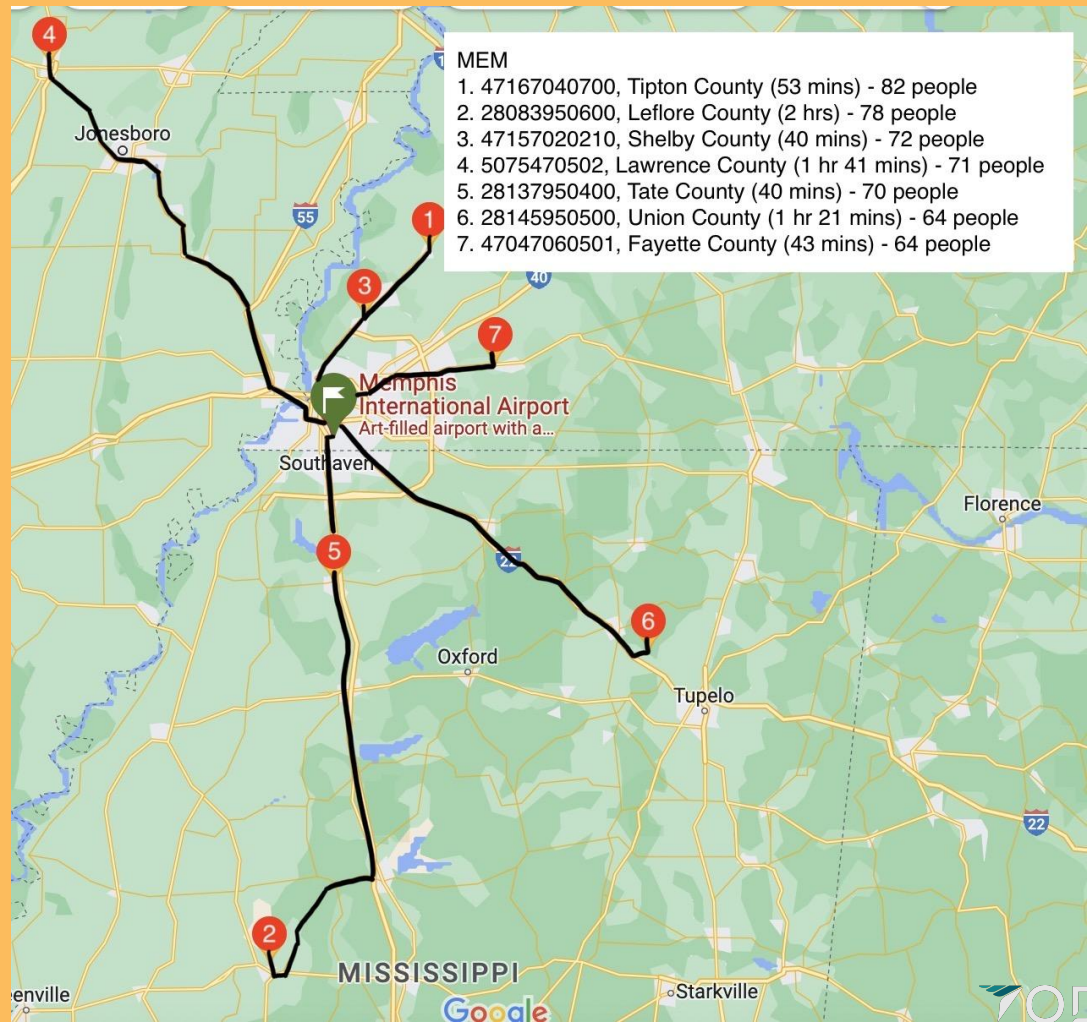


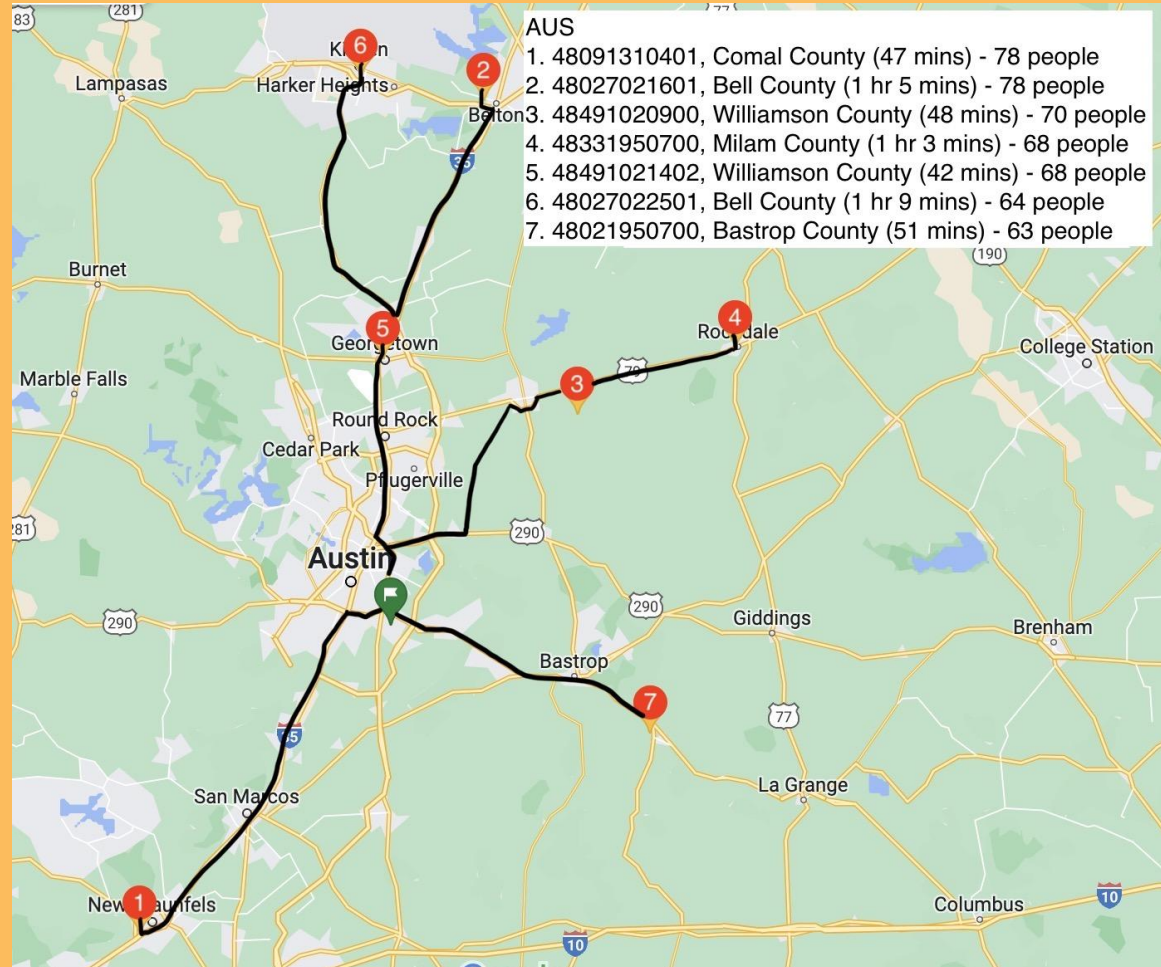
OKC

OKC

1. 40009966100, Beckham County (1 hr 42 mins) - 91 people
2. 40087400300, McClain County (40 mins) - 84 people
3. 40009966200, Beckham County (1 hr 50 mins) - 76 people
4. 40083600700, Logan County (55 mins) - 75 people
5. 40133583400, Seminole County (59 mins) - 74 people
6. 40017300201, Canadian County (42 mins) - 72 people
7. 40039960600, Custer County (1 hr 11 mins) - 68 people

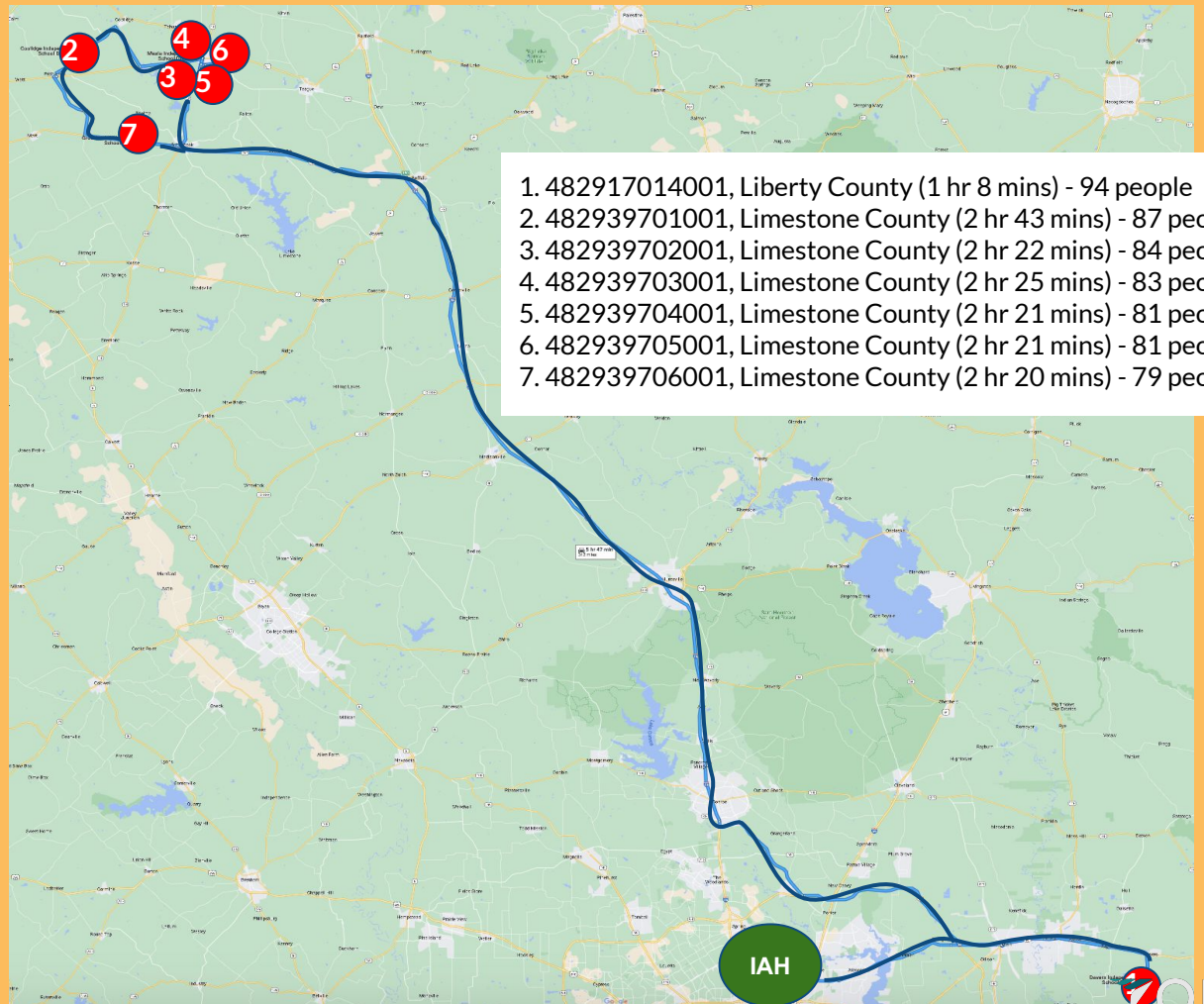


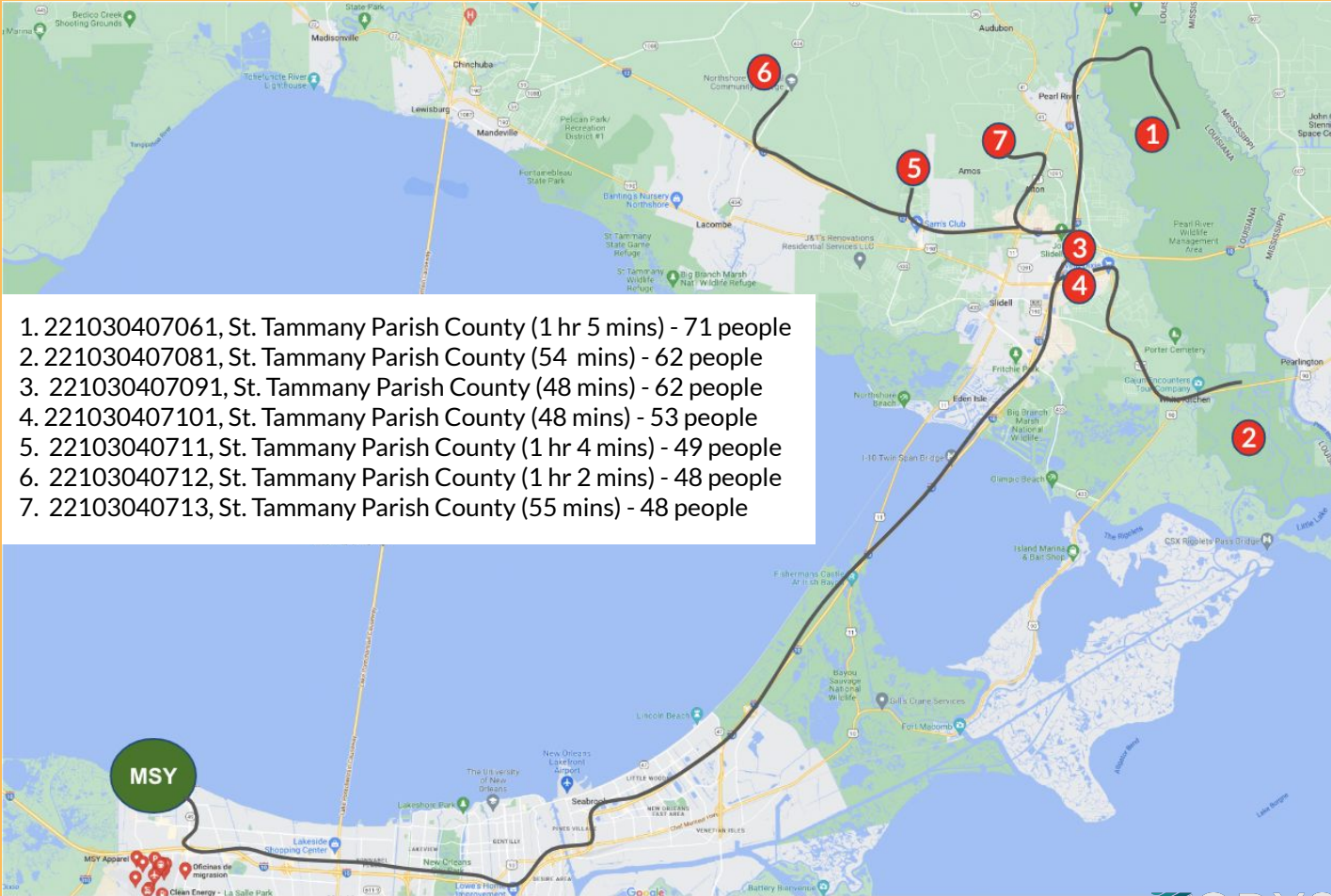


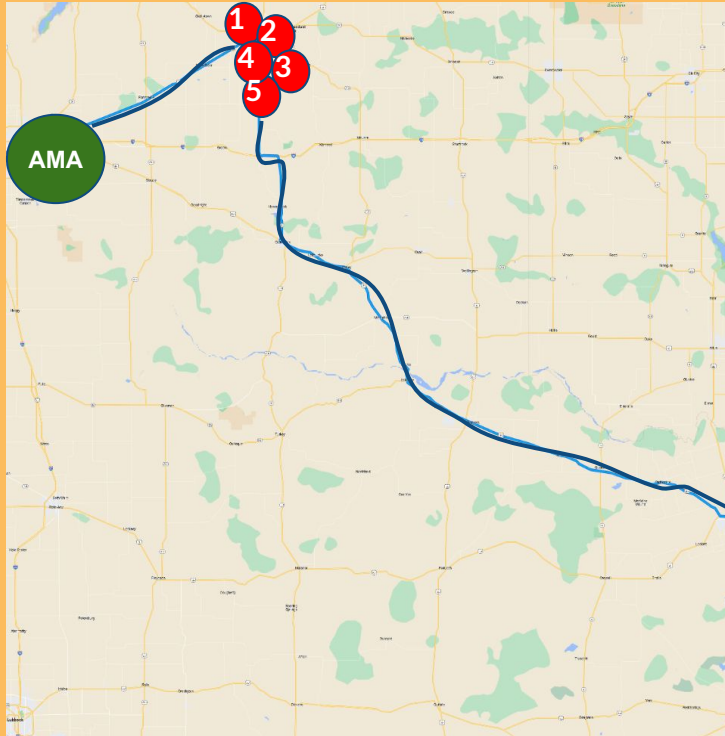




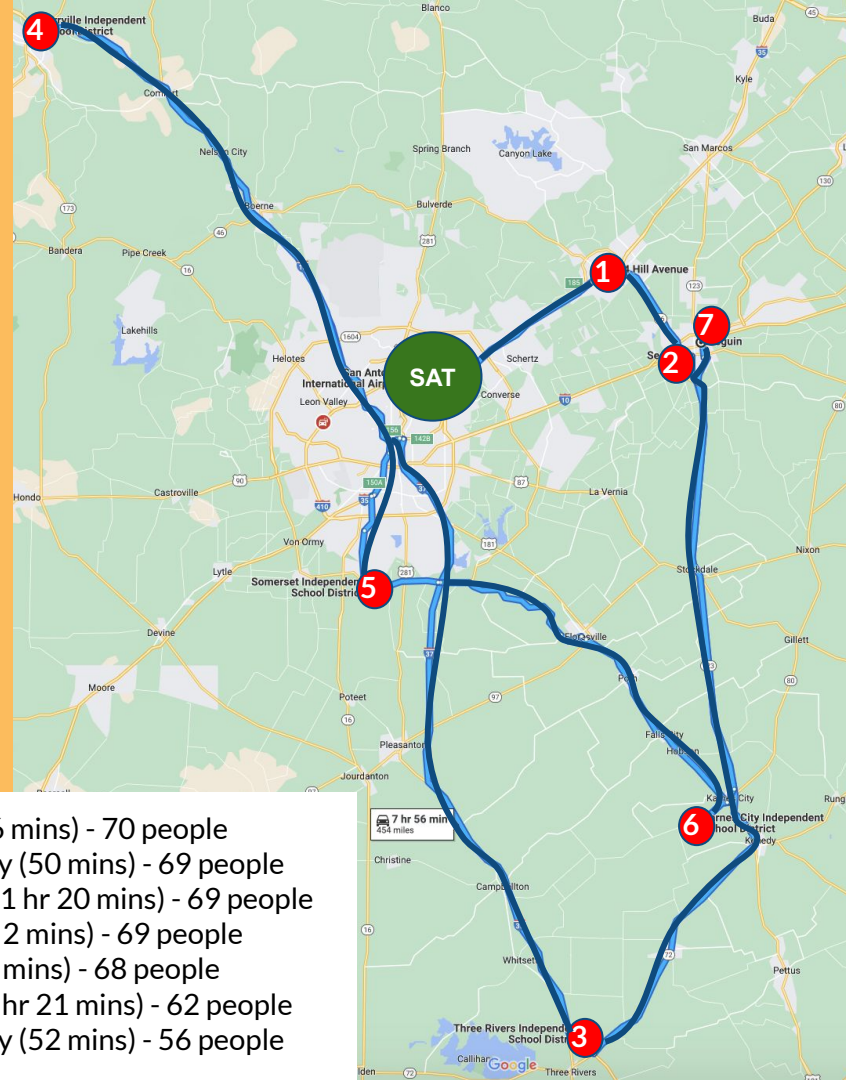
IAH



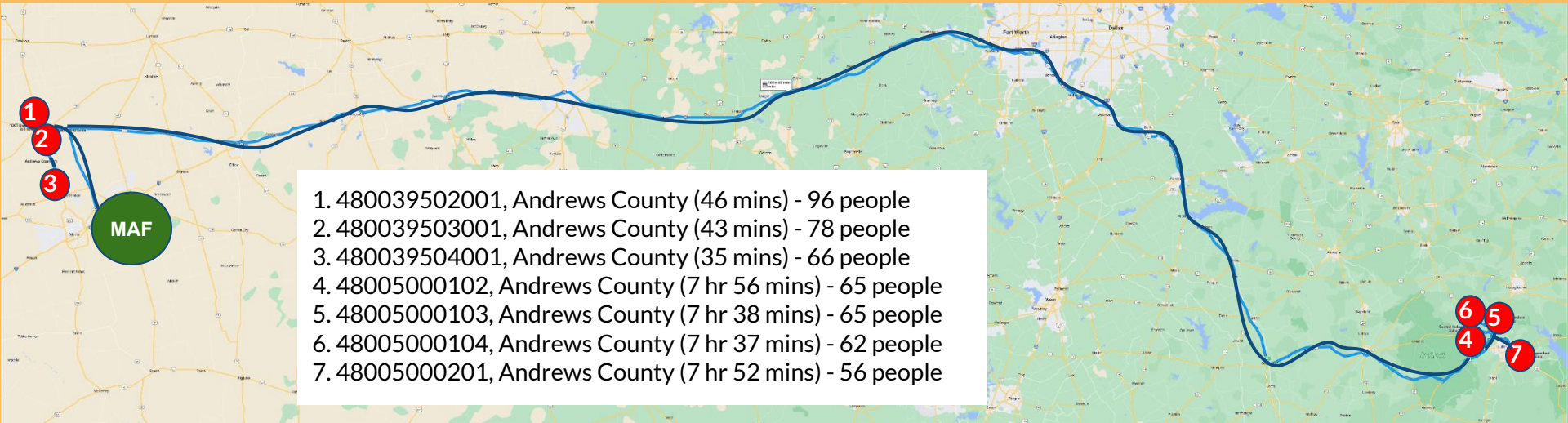


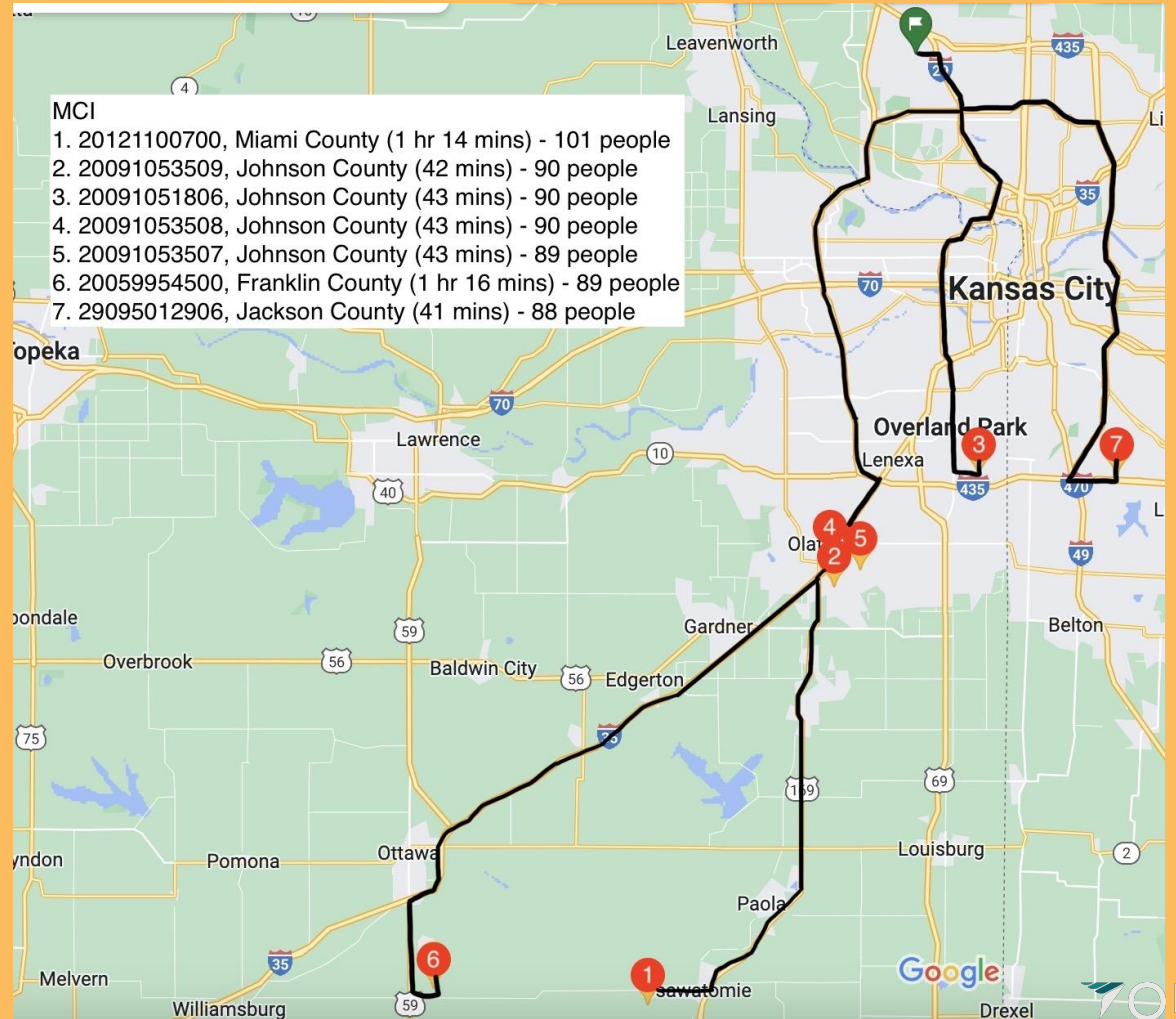


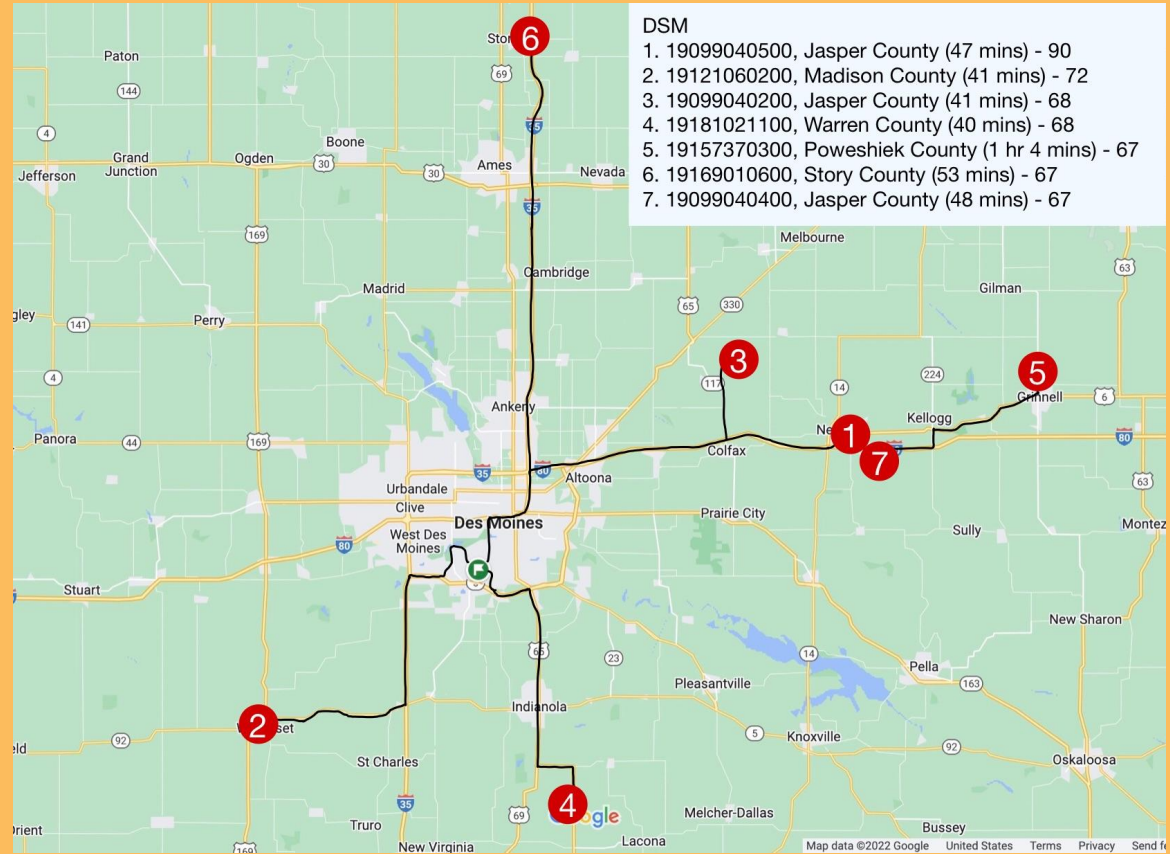
1. 481799504001, Gray County (54 mins) - 93 people
2. 481799505001 Gray County (55 mins) - 80 people
3. 481799506001, Gray County (53 mins) - 79 people
4. 481799507001, Gray County (50 mins) - 74 people
5. 481799508001, Gray County (51 mins) - 67 people
6. 48181000101, Gray County (5 hr 31 mins) - 66 people
7. 48181000102, Gray County (5 hr 33 mins) - 64 people



1. 48091310100, Comal County (36 mins) - 70 people
2. 48187210300, Guadalupe County (50 mins) - 69 people
3. 48297950100, Live Oak County (1 hr 20 mins) - 69 people
4. 48265960500, Kerr County (1 hr 2 mins) - 69 people
5. 48029152100, Bexar County (40 mins) - 68 people
6. 48255970200, Karnes County (1 hr 21 mins) - 62 people
7. 48187210100, Guadalupe County (52 mins) - 56 people







Relevant Functions

def topTracts

Input: a specific airport location

Return: a table with the most popular origin tracts and the number of drivers from each top tract

def topServicableTracts

Input: a specific airport location

Return: a table with the most popular origin tracts and the number of drivers from each top tract a given distance away from the airport of interest

def incomeDataSorting

Input: a specific airport location

Return: a table with the percentage of drivers that fall into the following income buckets: 0-50K, 50-100K, 100-150K, 150K+

Relevant Functions

def totalRevenue

Input: a hub airport, a list of appropriate local airports

Prints:

Number of flyers captured from P2P

Number of flyers captured from CE

P2P revenue

CE revenue

Total number of flyers

Total Revenue

Return: the total possible annual revenue from adding Odys eVTOLs

def p2p

Input: a hub airport

Return: the number of available drivers for P2P capture

def catchmentExpansion

Input: a catchment airport

Return: the number of available drivers for CE capture based on the number of long haul flyers out of the catchment airport + drivers whose drive time to the catchment airport is longer than the Odys flight time to the hub airport



Questions?

Sources

Past 12 month revenue values for each airline in 2022: [macrotrends.net](https://www.macrotrends.net)

<https://www.macrotrends.net/stocks/charts/DAL/delta-air-lines/revenue>

<https://www.macrotrends.net/stocks/charts/AAL/american-airlines-group/revenue>

<https://www.macrotrends.net/stocks/charts/UAL/united-airlines-holdings-inc/revenue>